CHAPTER

9 R&D-based endogenous growth

In all our growth models the ultimate source of growth in GDP per worker is technological progress. Up to now technological progress has been either *unexplained* or *unintentional*. In this chapter we will study a growth model in which *technological progress is the endogenous and deliberate outcome of a production process which requires productive inputs.*

The exogenous growth models of Part 2 postulated a fixed growth rate in the technological variable A_t, so the driving force of economic growth was basically *unexplained*. In the externality-based endogenous growth models presented in the previous chapter, A_t was linked to aggregate capital (or income), so growth in A_t was endogenous and 'explained' by growth in capital, but technological progress was an *unintentional* by-product of economic activity in general. No agent was concerned with creating technological progress, and no inputs were used for the purpose of producing increases in A_t.

In the real world many people are occupied with producing better technology and production methods in private companies as well as at private and public universities and other organizations engaged in the creation of new knowledge.

In this chapter we study growth models that bring research and development, R&D, production processes explicitly into the picture. If economic growth is rooted in technological progress, then by better understanding the production process for technological progress we gain a deeper understanding of growth.

9.1 The role of ideas in economic development

In our models technological progress will have a very precise meaning: there will still be a technological variable, A_t, and technological progress will mean that A_t increases. The ultimate output of the R&D sector in period t is thus the total technological progress, $A_{t+1} - A_t$, from period t to period $t+1$.

We view the production process in the R&D sector as follows. Economic resources in research and development (or in innovative effort in general) produce *ideas*. These ideas are what ultimately bring better technology, that is, increases in A_t. So, *better technology is created by new ideas*, where we interpret ideas in a broad sense. Ideas can be new insights in basic research, such as Newton's mechanics, calculus, Einstein's theory of relativism, Bohr's atomic theory, etc. They can also be more 'close-to-the-factory' inventions of new machinery, such as the steam engine or a new generation of computers, or better 'tools' for existing machinery, such as a new or improved piece of software for existing computers. Furthermore, they can be ideas for new and more productive ways to organize production, such as the assembly line, or they can be ideas for new intermediate or final products.

Ideas come from people, so the R&D production process has to be labour intensive. We assume that the only inputs in the R&D sector are the labour power of 'researchers' and the existing stock of knowledge, although in the real world also some capital is used in R&D activities. We thus view the R&D sector as follows:

$$\text{Inputs: the innovative effort of labour and the existing level of technology}$$
$$\xrightarrow{\text{create as}}$$
$$\text{Direct output: new insights and ideas}$$
$$\xrightarrow{\text{which result in the}}$$
$$\text{Ultimate output: a new and higher level of technology.}$$

The fact that the direct output of the R&D sector essentially consists of ideas makes the associated microeconomics more subtle than in the growth models we have considered earlier. The reason is that ideas, viewed as economic goods, are like *public goods*. They are *non-rival*: the fact that one person uses calculus or (the idea behind) a specific spreadsheet in no way limits another person's ability to use calculus or the same kind of spreadsheet. Ideas are also *imperfectly excludable*, varying from absolutely non-excludable (no person can be prevented from using calculus) to partly excludable (in principle people are excluded from using a spreadsheet they haven't paid for because using it without a licence is a violation of law, but pirate copying takes place).

The non-rival character of ideas implies that private production of ideas cannot take place under perfect competition. Imperfect competition is therefore a necessary condition for privately conducted R&D, but at the same time imperfect competition violates a basic condition for economic efficiency, namely the principle that consumer prices should equal the marginal costs of production. Furthermore, the imperfect excludability of ideas means that private for-profit production involving the creation of new ideas may not be a viable option at all. This is explained in Section 9.2 below.

The public goods nature of ideas thus gives reasons why the production of improved technology should be done by non-profit enterprises. Why then is not all production of ideas public in the real world? Certainly a lot of research and development takes place in government and non-profit organizations like universities and hospitals. This is particularly true for most basic research. However, a lot of R&D activities take place in private firms where the motive for creating new ideas is the profit that can be earned from the ideas being marketed and transformed into new technology. For instance, private medical companies have large and very expensive R&D departments struggling to create new, or improve upon existing, types of medicine. The motive is to earn profits from selling the new or improved medicine.

In capitalist market economies the profit motive is used as a device for the allocation of resources. If there is a shortage of potatoes the price goes up, which makes it more profitable to produce potatoes. Profit-seeking producers will therefore produce more, thereby working to eliminate the shortage. Each firm is motivated by its profit, but comes, through its attempts to maximize profits, to serve society, 'as if guided by an invisible hand' in Adam Smith's famous words. Most economists think that markets and the profit motive constitute a mechanism for the distribution of scarce resources over competing ends that is so effective that it should be widely used in actual resource allocation. (Most economists think as well that the market mechanism is not perfect and should be 'corrected' by various forms of government intervention to address imperfections such as externalities, public goods, abuse of market power, etc.)

If the market mechanism is useful in allocating resources between the production of potatoes and bread, say, it can perhaps also be effective in allocating resources to R&D activities. Private R&D firms create new ideas motivated by the profits that can be earned from selling the ideas, but thereby they come to act as technological developers to the benefit of everybody. Thus, society may have an interest in R&D activities being conducted privately in so far as the profit motive is considered an effective engine for technological development. However, the public goods nature of ideas implies difficulties for the working of this engine.

Section 9.2 explains and discusses the microeconomics of ideas and explains the difficulties of private production of new technology. It also describes some resolutions to these difficulties. From Section 9.3 an R&D-based growth model is presented. Since the microeconomics of (private) R&D production is too subtle for a textbook at this level, the model will abstract from these subtleties and be of a more conventional macroeconomic character. We simply lay out some relationships between technology and growth and study their implications. Although our model does not include an explicit treatment of the microeconomics of (private) production of ideas, this subtle microeconomics will be visible in the model, as we shall see.[1]

9.2 Ideas as economic goods

A private production of ideas requires that ideas can somehow create an income. In the real world income creation from ideas takes different forms. Sometimes an idea is invented by a pure R&D firm that does not intend to use the idea itself. The inventing firm will then want to sell the idea to another firm. This will most often take place by the selling of a *patent*, or by licensing (renting) out a patent in return for a royalty income paid period by period. In other cases ideas are created in research departments of firms that want to use the idea themselves in the production of commodities. The idea is then patented by the producing firm itself and creates an income through the selling of a good containing the idea. For instance, the main content in a new improved piece of software, like a new spreadsheet, is the idea of the computer code, but this is sold to the customer in a box containing a CD, a manual, etc. Likewise, the real novelty in a new generation of razors is the idea for an improvement (now with five blades and ball bearings), but the idea is sold packaged with a traditional good consisting of the razor itself and some blades. We may view the invention of an idea and its use in production as two separate activities. In any case, patents seem to play a crucial role in ensuring that ideas can generate income for those who developed them. This section explains why this is the case.

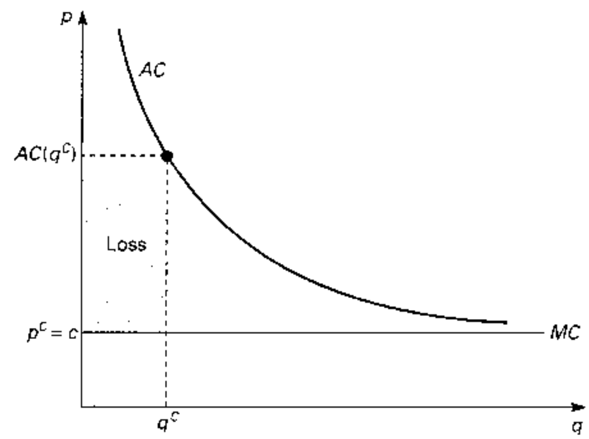
The non-rival nature of ideas

Once a new idea has been invented, the use of the idea in one activity does not in any way limit the extent to which it can be used in other activities. This feature gives ideas the nature of public goods. In relation to public goods, one important concept is *rivalry*. What we just said means that *ideas are non-rival*.

An economic good can be rival or non-rival. It is rival if the consumption or use of the good by one person or firm prevents other persons or firms from using it. Potatoes and bread, restaurant meals and cars are rival goods. A good is non-rival if its use by one agent does not limit the ability of other agents to use the same good. Classic examples of non-rival goods are national defence, the services of the legal system ('law and order'), the services of

1. The R&D-based growth model is known as the Romer model of endogenous growth and was developed by Paul M. Romer over several papers, in particular 'Endogenous Technological Change', *Journal of Political Economy*, **98**, 1990, pp. S71–S102. (We present a generalized version of the model due to Charles I. Jones, 'R&D-based Models of Economic Growth', *Journal of Political Economy*, **103**, 1995, pp. 759–784.) It is the full model, including the subtle microeconomics, that Romer has become famous for. Macro models close to the one we study in this chapter had been suggested before. One important conclusion of Romer's model with a true microeconomic foundation is that in a certain sense it implies the macro model we study in this chapter, making this much less *ad hoc* than it first appears.

FIGURE 9.1 Production of a research intensive product, perfect competition



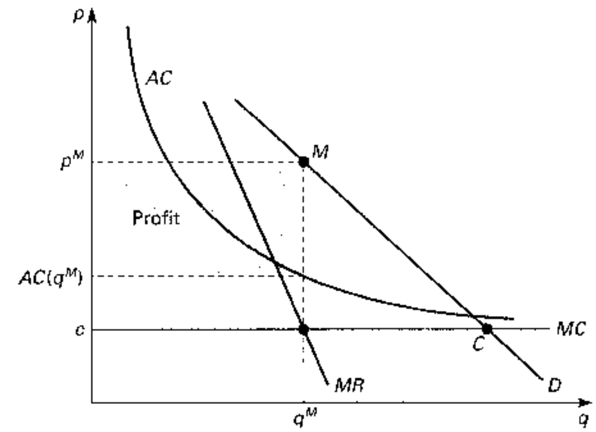
bridges and parks (in so far as they are not overcrowded), access to fresh air, great scenic views, etc. One aspect of a non-rival good is that its marginal cost of production is (close to) zero. One additional passage of a bridge, for instance, involves almost no additional cost (again provided it is not already congested), although the bridge may originally have been extremely costly to construct.

Ideas are non-rival. For instance, the big discoveries in basic research, like calculus, can be used by many different persons in many different places at the same time, and the extent to which it is used in one place at one time has no implication for the extent to which it can be used in other places at the same or at other times. The non-rival character of ideas has an important implication for the cost functions associated with the use of ideas, for instance in the production of a traditional good that embodies the idea.

Let us return to the example of a new generation of razors. The idea behind it has been developed either by an independent R&D firm or in the research department of the producing firm. In any case, acquiring the idea has given rise to a cost consisting mainly of the incomes of the developers. Once the idea has been fully developed, the use of the idea in one 'activity', the production of a specific razor unit, does not prevent the idea from being used in another activity, the production of another razor unit. Hence, the cost of acquiring the idea is to be considered as a fixed cost, F say, in the production of the new razors that carry the idea. Let us assume that in addition to this fixed cost there is a (small) constant unit cost, c, of producing razors, as would follow from the replication argument and given input prices (explain why). If we denote the firm's production of razors by q, the cost function is $C(q) = F + cq$, and hence the marginal and average cost functions, respectively, are $MC(q) = c$, and $AC(q) = F/q + c$. These are illustrated in Fig. 9.1.

Assume now that the producer is a small perfect competitor who takes the price of razors as given. The horizontal marginal cost curve will then also be the firm's supply curve: if the given razor price exceeds c, the supply will be infinite; if the output price is smaller than c, supply will be zero, and if the razor price is exactly c, any output level is an optimal supply. Once the idea has been fully developed, the development cost is 'sunk', and therefore optimal behaviour only depends on the variable cost component, c. In these circumstances market forces should imply a competitive equilibrium where the price, p^C, is equal to c, and the amount produced is given by demand, for instance for the small firm implying a production of q^C as indicated in Fig. 9.1. In this equilibrium the firm will earn a negative profit since average cost, $AC(q^C)$, exceeds the price, $p^C = c$. The total revenue p^Cq^C exactly

FIGURE 9.2 Production of a research intensive product, monopoly



covers the *variable* cost cq^C, but falls short of total cost by the amount $(AC(q^C) - c)q^C = F$. The fixed cost is uncovered and the production of razors involves a loss of F.

However, before the firm decides to develop or buy the idea behind the new razor it would be able to foresee what we have just explained: that the whole project can only end up yielding negative profits. The firm would therefore never decide to start such a project in the first place. The conclusion is that *under the conditions of perfect competition (price taking in the output market), one cannot have a private production of idea-based technological progress.*

Assume now instead that the firm is not small in the output market, but rather is the only producer of the particular new razor. Let the demand curve for this razor be as illustrated by D in Fig. 9.2. The cost curves are the same as in Fig. 9.1. With this demand curve the competitive equilibrium would be given by the point C involving the price c and a specific total output (of which we imagined the small firm above to supply q^C). When the firm is the sole supplier, it can take advantage of the fact that reducing output from the competitive level will imply a higher price (while the competitive firm's supply was assumed to be so small relative to the market that it had no significant influence on the price). It is well known from microeconomics that the monopoly firm should reduce output as long as marginal cost is larger than marginal revenue. This leads to the optimal point M in Fig. 9.2, where the marginal cost curve, MC, and the marginal revenue curve, MR, intersect. At the monopoly optimum, the price, p^M, could well be above average cost as illustrated, and the project of developing the idea and producing the new type of razor could thus imply positive profits. *Under conditions of imperfect competition (monopoly power in the output market) it is possible to have a private production of idea-based technological progress.*

The conclusion is that *in order to have a private production of idea-based technological progress it must somehow be ensured that the inventor of an idea holds monopoly over the use of the idea*, or in the production of a good that embodies the idea, at least for some period. This is indeed an important conclusion, and patent rights and other legal ways of protecting intellectual property rights serve as instruments for creating such monopoly power. Before turning to that, however, we should emphasize that this way of ensuring a profit-motivated development of ideas is highly imperfect, exactly because it works through giving monopoly power to the firm.

In the monopoly optimum given by point M in Fig. 9.2, the price is above marginal cost (otherwise there would not be positive profits). This means that the consumers are ready to pay more for additional units of the new razor than the additional cost that would result from producing additional units. Hence, from a social viewpoint, too few resources are devoted to the utilization of the idea behind the new type of razors: Monopoly implies a deadweight loss in the form of a foregone consumers' surplus, in the figure represented by the area of the triangle below the D-curve, above the MC-curve, and to the right of the vertical line at q^M. In other words, *if one ensures a private production of idea-based technological progress by giving monopoly in the use of ideas to the creators, then the resulting ideas will be used too little as compared to a social optimum.*

This might seem to suggest that research and development should mainly be conducted by public, or other non-profit, enterprises. These could supply ideas, or products embodying ideas, at marginal cost, thus ensuring an optimal utilization of the ideas and simply accept the implied negative profits. The losses would, however, have to be financed by taxes, and taxes have negative welfare effects too because they imply distortions. The social cost of having a less than optimal utilization of ideas under private R&D would have to be weighed against the social cost of distortionary taxation under public R&D. Furthermore, as mentioned earlier, we (the general public) may wish to have private R&D activities, and be willing to live with the monopoly power necessarily associated with it, in so far as we believe that sometimes the profit motive is an effective engine for technological development. The experiences of the formerly planned economies of Eastern Europe suggest that the more applied part of research and development is done most effectively in private enterprises. When it comes to developing the kind of razor, toothbrush, or blue jeans that people really want, or the kind of spreadsheet or catalyser that other firms can really use, private developers are, probably, more effective than public bureaucracies exactly because they are governed by market signals and the profit motive.

Private R&D activities may therefore be socially beneficial, and certainly in the real world we do observe a lot of R&D taking place in private firms. This state of affairs requires, as we have seen, that somehow the inventor of an idea obtains monopoly power over the use of the idea. Very often patents serve the purpose of ensuring such a monopoly. When Novo Nordisk, at huge development costs, has designed a new way of producing an improved form of insulin, it can protect its 'design' by taking out a patent on it. The patent gives Novo Nordisk the right to be the sole user of the design in production of insulin for some period. If another firm uses the same design in the production of insulin, it will violate the law. What the patent really does is thus to exclude other persons and firms from the use of the idea for a specific purpose. This leads us naturally to the discussion of another essential concept in relation to public goods, namely *excludability.*

The importance of excludability

A good is excludable if the owner of the good can exclude other persons or firms from using it (if they do not pay for it). Most traditional goods, like bread and potatoes, haircuts, restaurant meals and travels, etc., are excludable. They are only handed over the desk upon payment. A good is non-excludable if persons or firms cannot be excluded from the use of it. Classic examples of non-excludable goods are, again, national defence, 'law and order', clean air, etc. Since the access to a non-excludable good cannot be made conditional on payment, non-excludable goods cannot be produced and marketed by private, profit-motivated firms. Only public enterprises not motivated by profits can produce completely non-excludable goods.

Most economic goods are either both rival and excludable, or both non-rival and non-excludable. The first are the traditional goods like bread, potatoes, haircuts, restaurant meals, etc., and the latter are the pure public goods like national defence, law and order, fresh air, etc. However, rivalry and excludability are conceptually distinct. There are goods that are non-rival and excludable, for instance crossing a (not congested) bridge. One person's passage will (under normal circumstances) not limit other persons' possibilities of crossing the same bridge, so the good is non-rival. People can, however, be excluded from crossing a bridge if they do not pay, so the good is excludable. An example of a good that is rival, but non-excludable could be 'fish in the sea', since a fish caught by one person cannot be caught by another person too, but during most of history free access to catching fish in the open sea has been considered a right for everybody.

While a good is either rival or non-rival (again disregarding congestion), excludability can occur to different degrees. Furthermore, and relatedly, increased excludability can to some extent be ensured by law and by technology. These two features are important in connection with ideas.

Consider again the idea for a new way of producing an improved form of insulin. Each package of medicine will somehow contain the idea, but the exact recipe will not be directly visible and hence not completely open for use by everybody. The idea or design is therefore, by nature, somewhat excludable. On the other hand, a competitor with a research department of its own could probably, by analysing the product, break or read the code and in that way technically become able to produce the same medicine. Hence the design is not perfectly excludable.

The legal institution of patent rights can to some extent prevent direct copying and thus make the design more excludable than it was by nature. There may also be ways to produce the new medicine that can make it more difficult to detect the invention behind it, just like a TV transmission in the air (another non-rival good) can be made more excludable by encoding the signal. Both legal and technological devices can serve to make ideas more excludable.

Note carefully what a patent does, for example in connection with a new design for insulin: it prevents a competitor who can break the code from using the same design for the production of insulin. It does not prevent a competitor from analysing the product or from studying the application for the patent, thereby getting full insight into the design. In this way a competitor may obtain increased knowledge that can be used for a related idea not protected by the patent. The knowledge first created by Novo Nordisk can be useful for other developers even though the patent prevents them from using this knowledge for exactly the same purpose as Novo Nordisk.

While non-rivalry prevents private idea-based production under perfect competition, complete non-excludability prevents private production under any market structure. If one cannot prevent people who do not pay for a good from using it, no money can be made from producing the good. At least some degree of excludability is needed in order to have private production.

By nature, ideas vary between being absolutely non-excludable and being imperfectly excludable. Discoveries in basic research are non-excludable, while more down-to-earth ideas for new products or production methods most often are partly excludable: they can be copied, but at some cost. In the latter case, the degree of excludability can often be much increased by legal or technological arrangements.

For these reasons almost all basic research takes place at public universities or at other types of non-profit research institutions. On the other hand, a lot of R&D directed at creating new products and improved production methods takes place in private firms. The latter requires some degree of excludability, which is often ensured by the presence of a legal system protecting intellectual property rights. At the same time, this legal system, by giving (through a patent) the exclusive right to use a certain idea for a certain purpose to the inventor, creates the monopoly power in the use of the idea that is necessary to overcome the problem associated with the non-rival character of ideas. *A legal system of protection of intellectual property rights thus serves the double purpose of ensuring some excludability and ensuring some monopoly power in the use of new ideas.*

THE MICROECONOMICS OF PRIVATE PRODUCTION OF IDEAS

The non-rival character of ideas implies that idea-based technological progress driven by private R&D will only take place if the inventor of an idea holds monopoly over its use. This monopoly power means that ideas, once developed, will be used less than what is socially optimal.

The imperfect excludability of ideas threatens the commercial viability of private R&D, but legal and technical arrangements can increase excludability.

A legal system for the protection of intellectual property rights (patents) can work to ensure both a relatively large degree of excludability and the monopoly power required for a private production of ideas.

The role of intellectual property rights in economic development

Some economic historians, for instance Nobel Prize winner Douglass C. North, consider this point to be absolutely essential for understanding the unprecedented economic growth over the past 200 years. As we have touched upon before, the Western world has experienced average annual growth rates of GDP per capita of about 2 per cent during the past 200 years, while over the several thousands of years before that, the average annual growth rate of GDP per capita must have been close to zero to fit with the level of income per capita around year 1800. Apparently big changes must have occurred around the late eighteenth and early nineteenth centuries, changes that have worked to initiate and maintain technological progress and economic growth.

What kind of an event could that be? According to some (Marxian, for instance) historians the big event was the change in the mode of production from feudalism to capitalism. During centuries of feudalism a very slow technological evolution had taken place. Gradually during this process, technology came to be in increasing contradiction to the social relations of feudalism characterized by labour power being tied to rural estates and lots of rules and regulations preventing the free initiative to establish new firms in trade and industry. A social revolution had to occur, and the revolution that took place was the transition from the feudal to the capitalist mode of production. Among other features, this transition implied that labour power was set free from (or kicked out of) rural estates, thus obtaining 'freedom' to move to the cities looking for work in the growing manufacturing sector. Another feature was the abolition of rules and regulations that had hitherto prevented free enterprise. In this way the profit motive and the allocation of resources through markets became much more important than it had been earlier. The market mechanism thus came to work as a driving force for the creation of new technologies. Each firm, competing with other firms, would have an individual incentive to create new production methods that could increase productivity and reduce costs. After any such invention there would be a period where the competitors had not yet adopted the new technique. The output price would therefore, during this period, be determined by the higher costs of the competing firms, and the inventing firm would temporarily earn above-normal profits. In this way the profit motive would continuously work as an incentive for further technological progress.

This story certainly sounds appealing, but according to the economic historians mentioned above, one element is missing. If new inventions were in no way protected from being copied, then the time that an inventor would be alone with a new production method would be short, so there would not be much extra profit to reap. This state of affairs would prevent inventing in the first place (note that this argument is basically the same as our argument that there can be no private idea-based technological progress under perfect competition). The important feature missing from the Marxian story, still according to these economic historians, was an improvement in the legal protection of intellectual property rights that had occurred gradually up to the time we are talking about, at that time reaching the 'critical' level. By protecting through patents new ideas from being copied, a private return on the effort exerted in developing ideas was ensured. This may have improved greatly the incentives for a private production of ideas that in turn contributed to the technological progress witnessed over the past 200 years.

With these remarks in mind it seems worthwhile to set up a formal model which focuses on research and development as a source of technological evolution and growth. As mentioned, our model will be of a macroeconomic nature showing only some signs of the subtle microeconomics of the production of ideas.

9.3 An R&D-based macro growth model

The model we study in this chapter is meant to explain the relatively constant economic growth of around 1.5–2 per cent per year in the developed world, *considering the developed world as one large economy*. Since the economy we consider will produce all of its technological progress itself, and thus does not import technology developed elsewhere, it is implicit that the model covers the entire developed world.

There will be two types of output: 'new technology', and final goods that can be used for consumption or for investment in physical capital. We will now describe the production processes for each type of output.

The production of final goods

A representative firm produces final output in the amount Y_t according to the familiar Cobb–Douglas production function:

$$Y_t = K_t^{\alpha}(A_t L_{Yt})^{1-\alpha}, \qquad 0 < \alpha < 1. \tag{1}$$

Formally, the only new feature is that labour input is now denoted by L_{Yt} rather than by L_t. This is because there will also be a labour input, L_{At}, in the R&D sector and we have to distinguish between the two. Another input is capital services, the amount of which we assume to be proportional to the stock of capital, K_t. Finally the technological variable, A_t, enters as usual.

From now we will think of A_t more as an input in line with K_t and L_{Yt}. The variable A_t is the total productive effect of the stock of all innovative ideas that have come into existence up to period t. There are important differences between the input A_t and the inputs K_t and L_{Yt}: first, the representative firm cannot adjust its input of technology as it wishes. From the point of view of each individual firm, society's stock of ideas or technology, A_t, is given. Second, the fixed amount of technology, A_t, can be used in every firm because the general level of technology is non-rival.

According to the replication argument, the production function should have constant returns to the two *rival* inputs, K_t and L_{Yt}, as assumed in (1). Doubling production by undertaking the same activities twice means doubling the inputs of capital and labour *using one and the same input of technology twice*. As a consequence, the production function in (1) has *increasing returns* in the three inputs, K_t, L_{Yt} and A_t, a feature that is important for our conclusions.

The production of new technology

The R&D sector also has one representative firm. As usual we should distinguish between its two roles. On the one hand, it produces the entire output, $A_{t+1} - A_t$, of new technology in period t, since there is only this one firm. On the other hand, the firm is actually to be considered as one of many small firms and therefore it has only a negligible influence on the aggregates of the R&D sector, that is, on the total input of labour, L_{At}, and on the total output, $A_{t+1} - A_t$, produced in the sector in period t. Consequently, the representative firm in the R&D sector takes the stock of technology as given at any point in time, just like the firm in the final goods sector does.

The production function of the individual firm is:

$$a_t = \bar{\rho} A_t^{\phi} L_{At}^{d}, \qquad \bar{\rho} > 0, \tag{2}$$

where a_t is the number of innovative ideas produced by the individual firm, and L_{At}^{d} is the labour demand of the individual firm in period t. In equilibrium a_t and L_{At}^{d} have to equal the total production of ideas, $A_{t+1} - A_t$, and the total labour input, L_{At}, respectively.

As already explained, we make the simplifying assumption that capital is not required as an input in the R&D sector. Hence the production function should, according to the replication argument, have constant returns to the input of labour alone.[2]

The existing stock of knowledge, A_t, is an input for the R&D firms, just as it is for the producers of final output, its output elasticity being ϕ, and like the firms in the final goods sector, each individual firm in the R&D sector takes A_t as given. Just as existing insights, production methods, computer software, etc., are productive in the final goods sector, they are productive in research and development. Hence, existing technology may promote the production of new technology, an effect sometimes referred to as 'standing on shoulders'. This effect speaks for a positive ϕ, possibly equal to 1. On the other hand, it may be that the more ideas that have already been accumulated in the existing stock of knowledge, A_t, the more difficult it becomes to get new ideas. This 'fishing out' effect is based on the view that there is a given, large (possibly infinite) pool of potential ideas and the easiest ones are discovered first, so the more ideas that have already been harvested, the more difficult it becomes to find a new one. This effect speaks for a negative value of ϕ. For the time being we will not impose any restrictions on the size or sign of ϕ, but generally we will think of ϕ as positive.

At the aggregate level we will assume that there may be a negative externality from the aggregate use of labour, L_{At}, in the R&D sector:

$$\bar{\rho} = \rho L_{At}^{\lambda - 1}, \qquad \rho > 0,\ 0 < \lambda \le 1. \tag{3}$$

If $\lambda < 1$, then there is a negative spillover from the aggregate activity level in the R&D sector to the productivity of the individual firm, while if $\lambda = 1$, there is no such effect. The motivation for the possibility of such an effect is 'stepping on toes': the more research there is in the R&D sector, the more probable it is that several individual firms work with the same ideas. Two firms creating the same idea only count for one in the stock of knowledge, A_t.

In equilibrium one must have $a_t = A_{t+1} - A_t$, and $L_{At}^{d} = L_{At}$, so from (2) and (3) the aggregate production function for new technology is:

$$A_{t+1} - A_t = \rho A_t^{\phi} L_{At}^{\lambda}. \tag{4}$$

Note some particular cases. If there is no negative 'stepping on toes' effect, $\lambda = 1$, then also at the level of the whole R&D sector, the number of ideas produced is proportional to the labour input in the sector. If the combination of the 'standing on shoulders' effect and

2. We are abstracting from an important aspect of real-life R&D, namely the stochastic nature of the arrival of new ideas. By doubling labour inputs, an R&D firm cannot be sure of doubling the number of ideas it will produce *in a deterministic sense* since ideas arrive in a less predictable way. Rather, a doubling of labour inputs could double the stochastic arrival intensity of ideas (in a Poisson process describing the production of ideas). An explicit modelling of stochastic arrival of ideas would not change our results qualitatively.

the 'fishing out' effect is such that $\phi = 1$, then (4) gives the growth rate of technology as: $(A_{t+1} - A_t)/A_t = \rho L_{At}^{\lambda}$.

The complete model

Our complete model consists of the six equations below. The first two are the aggregate production functions for the final goods sector and the R&D sector, respectively. The third one is the usual capital accumulation equation assuming an exogenous savings (investment) rate, s, and an exogenous depreciation rate, δ. Both s and δ are between 0 and 1. Likewise, (8) is the familiar equation assuming that the labour force, L_t, grows at a fixed exogenous rate, n, where $n > -1$. The two last equations are new. Equation (9) says that the sum of the labour inputs in the two sectors has to be equal to the total amount of labour available. Equation (10) assumes that in all periods a given and exogenous fraction, s_R, of all labour is used in the R&D sector, where $0 < s_R < 1$.

THE COMPLETE R&D-BASED GROWTH MODEL

$$Y_t = K_t^{\alpha}(A_t L_{Yt})^{1-\alpha}, \tag{5}$$

$$A_{t+1} - A_t = \rho A_t^{\phi} L_{At}^{\lambda}, \quad A_0 \text{ given} \tag{6}$$

$$K_{t+1} = sY_t + (1-\delta)K_t, \quad K_0 \text{ given} \tag{7}$$

$$L_{t+1} = (1+n)L_t, \quad L_0 \text{ given} \tag{8}$$

$$L_{Yt} + L_{At} = L_t \tag{9}$$

$$L_{At} = s_R L_t. \tag{10}$$

The parameter s_R, referred to as the R&D share, should be thought of as a behavioural parameter in line with s and n, while α, ρ, ϕ, λ and δ are technical parameters.

The state variables of the above model are K_t, L_t and A_t. For given initial values K_0, L_0 and A_0, the model will determine the full dynamic evolutions of all endogenous variables as illustrated in Fig. 9.3.

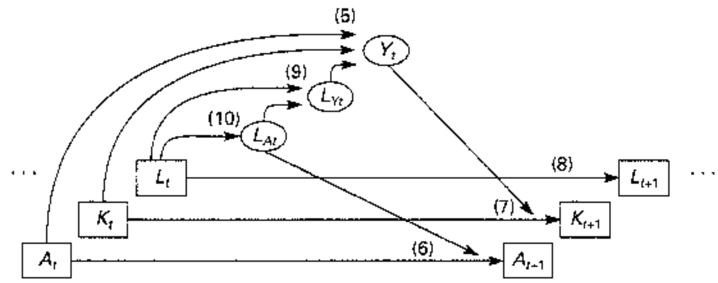
Because our model does not have an explicit micro foundation, we will have to treat the R&D share as exogenous and look into the consequences, rather than the causes, of its size. Thus, taking s_R as a given parameter, how large should we typically expect it to be, and how does it seem to change over time? Table 9.1 reports on the so-called GERD (gross expenditure on research and development), as a percentage of GDP in some OECD countries.

Typically, R&D shares in OECD countries seem to be around 1–4 per cent in the year 2005, with a tendency to have increased over the period from 1981 to 2005. Averaging over the 18 OECD countries of the table, the R&D share has increased from 2.0 to 2.4 per cent over the period corresponding to an average annual rate of increase in the R&D share of 0.8 per cent. As a crude extrapolation this corresponds to the R&D share roughly doubling every 100 years.

On the suppressed microeconomics of the model

As mentioned, we do not deal explicitly here with the microeconomics of our growth model. However, the microeconomics and the microeconomic problems of the model are visible in certain respects as it now stands. For instance, note that labour inputs used in

FIGURE 9.3 The dynamics of the R&D-based macro model



Note: Predetermined endogenous variables in squares, endogenous variables that can adjust in the period in circles.

TABLE 9.1 Gross expenditure on research and development (GERD) as a percentage of GDP, 18 OECD countries

Country	1981	2005	Average annual growth rate of GERD/GDP 1981–2005 (percent)
Austria	1.1	2.4	3.3
Canada	1.2	2.0	2.0
Denmark	1.0	2.4	3.6
Finland	1.2	3.5	4.7
France	1.9	2.1	0.5
Germany	2.2	2.5	0.6
Greece	0.2	0.6	5.7
Iceland	0.6	2.8	6.3
Ireland	0.7	1.3	2.7
Italy	0.9	1.1	1.0
Japan	2.3	3.3	1.5
Netherlands	1.8	1.7	−0.1
New Zealand	1.0	1.2	0.7
Norway	1.2	1.5	1.1
Spain	0.4	1.1	4.4
Sweden	2.2	3.8	2.3
UK	2.4	1.8	−1.2
USA	2.3	2.6	0.5
OECD average weighted by GDP in 1990	2.0	2.4	0.8

Source: OECD Science & Technology Data Base, 2007.

one sector cannot be used in the other, reflecting that labour power is a rival good, while the stock of knowledge can be used by all firms in both sectors simultaneously, reflecting that knowledge is non-rival. Furthermore, there is no equation for a factor reward rate in the model. In a more elaborate version of the model with an explicit description of its microeconomics, the wage rates and the other endogenous variables would adjust to ensure that real wages and the marginal products of labour are equalized across the two sectors.

You might also ask what the appropriate measure of GDP and national income in this model should be. There are two production sectors, so GDP will have to be some aggregate of the productions, Y_t and $A_{t+1} - A_t$, respectively, of the two sectors. Should it not be this aggregate income rather than just the Y_t that entered into the savings function? You may think of it as follows. The R&D sector is run by the government and its output is not marketed, but simply made freely available to everybody. People working in the R&D sector earn incomes, of course, but the government pays these incomes and finances its expenditure by taxing all income, so the total tax revenue equals the labour cost of running the research sector. Hence, total income after tax is the sum of the (values of the) productions in the two sectors minus the production in the R&D sector, that is, the Y_t of the final goods sector. This after-tax income enters into the savings function.

With respect to welfare, we will take the view that production of new technology does not create welfare in itself. It is only indirectly, as a means of increasing production of final goods, that technology creates welfare. Hence our measure of the prosperity of a society should (still) be the production of final goods per person, $Y_t/L_t \equiv y_t$, or rather the consumption of final goods per person, $c_t = (1 - s)y_t$.

9.4 The law of motion

We can describe the dynamics of the model by two difference equations that we now derive.

The dynamics of the rate of growth of technology

From (6) we can compute the growth rate in technology in any period by dividing both sides by A_t, and we can then, using (10), insert $s_R L_t$ for the L_{At} on the right-hand side:

$$g_t \equiv \frac{A_{t+1} - A_t}{A_t} = \rho A_t^{\phi-1} L_{At}^{\lambda} = \rho A_t^{\phi-1} (s_R L_t)^{\lambda}. \tag{11}$$

This shows that g_t is strictly positive in all periods. As long as just some labour input is used in the research sector there will be positive additions to knowledge all the time. If we write (11) for period $t + 1$ as well as period t and divide the first equation by the second, we get (with s_R constant):

$$\frac{g_{t+1}}{g_t} = \left(\frac{A_{t+1}}{A_t}\right)^{\phi-1} \left(\frac{L_{t+1}}{L_t}\right)^{\lambda}.$$

Here A_{t+1}/A_t is the growth *factor* of A_t, equal to the growth *rate* plus 1: $A_{t+1}/A_t = (A_{t+1} - A_t)/A_t + 1 = g_t + 1$. If we insert $(1 + g_t)$ for A_{t+1}/A_t, and insert $1 + n$ for L_{t+1}/L_t, we get:

$$\frac{g_{t+1}}{g_t} = (1 + g_t)^{\phi-1}(1 + n)^{\lambda},$$

which we can write as:

$$g_{t+1} = (1 + n)^{\lambda} g_t (1 + g_t)^{\phi-1}. \tag{12}$$

This is a first-order difference, or transition, equation in g_t alone. It has been derived exclusively from the three model equations that concern the research sector, (6), (8) and (10). A first conclusion from our model is thus that the growth rate, g_t, of technology has a dynamic evolution of its own: from any given initial value it develops independently of the other endogenous variables in a way that only depends on the parameters ϕ, λ and n (as long as s_R is constant).

The dynamic system also involving the economic variables

Let us define the usual auxiliary variables, $\tilde{k}_t \equiv K_t/(A_tL_t) = k_t/A_t$ and $\tilde{y}_t \equiv Y_t/(A_tL_t) = y_t/A_t$. Using (5), (9) and (10), we can write the production function as $Y_t = K_t^{\alpha}(A_t(1-s_R)L_t)^{1-\alpha}$. Dividing both sides by A_tL_t gives $\tilde{y}_t = \tilde{k}_t^{\alpha}(1-s_R)^{1-\alpha}$. From the capital accumulation equation, (7), dividing both sides by $A_{t+1}L_{t+1}$, we then get:

$$\tilde{k}_{t+1} = \frac{1}{(1+n)(1+g_t)}[s\tilde{k}_t^{\alpha}(1-s_R)^{1-\alpha} + (1-\delta)\tilde{k}_t]. \tag{13}$$

This has been derived in the same way that we derived transition equations in models considered earlier, but (13) is not a law of motion for $\tilde{k}_t$ since it has the endogenous g_t on the right-hand side. However, the system consisting of (12) and (13) is a law of motion in g_t and $\tilde{k}_t$. The two difference equations are only coupled in one direction: to compute $\tilde{k}_{t-1}$ from (13), one must know both $\tilde{k}_t$ and g_t, but to compute g_{t+1} from (12), one only needs to know g_t. This makes the dynamic system relatively easy to analyse.

Technological dynamics for $\phi = 1$ and for $0 < \phi < 1$

It will be highly illustrative and indicative for what goes on in this chapter to focus for the moment on the independent dynamics of technology, assuming a constant labour input in the research sector. Thus, assume that L_{At} is a constant, $L_A > 0$. In this case one gets from (11) the following growth rate of technology:

$$g_t = \rho A_t^{\phi-1} L_A^{\lambda}. \tag{14}$$

If $\phi = 1$, the technological growth rate, g_t, will be constant and positive and equal to ρL_A^{λ} in all periods. This implies that A_t will be ever increasing and go to infinity in the long run.

If $0 < \phi < 1$, the stock of knowledge, A_t, will also be increasing in all periods and go to infinity: in period t, the absolute change in technology is given directly by (6), $A_{t+1} - A_t = \rho A_t^{\phi} L_A^{\lambda}$, which is positive, and as A_t becomes larger and larger, the increase, $A_{t+1} - A_t$, in technology will also become larger and larger, implying that eventually A_t goes to infinity. This time, however, the *growth rate* of technology will decrease over time and eventually go to 0. This is easy to see from (14): as A_t goes to infinity, $A_t^{\phi-1}$ goes to 0 (remember $\phi < 1$). There will be increases in knowledge all the time, but when $\phi < 1$, the increases, $A_{t+1} - A_t$, cannot keep pace with the stock, A_t, because the existing stock of knowledge is not sufficiently productive in the creation of new knowledge. Hence, the growth rate of knowledge, $(A_{t+1} - A_t)/A_t$, has to fall.

It makes an important difference whether a fixed input of labour into the research sector creates a constant positive growth rate in technology ($\phi = 1$), or just an ever-increasing stock of knowledge with the growth rate of this going to 0 ($0 < \phi < 1$).

9.5 Semi-endogenous growth

In this section we consider $0 < \phi < 1$. For reasons that will become obvious we will, in this case, also assume that there is some growth in the labour force, $n > 0$.

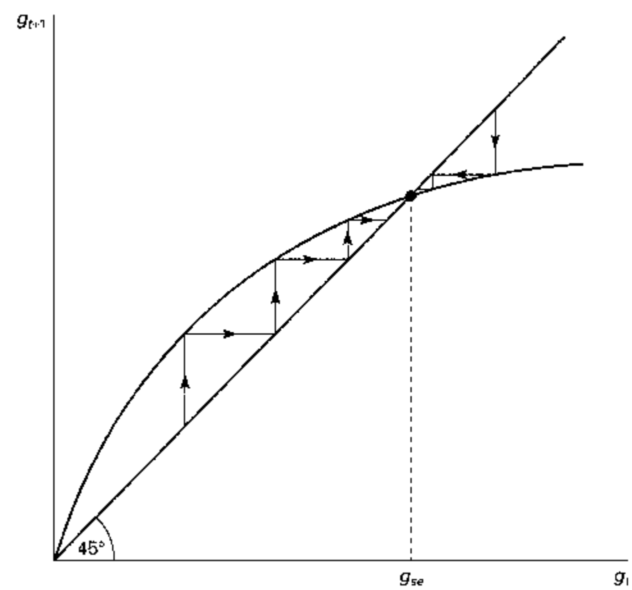
Convergence to steady state

Let us repeat the dynamic system we have arrived at:

$$g_{t+1} = (1+n)^{\lambda} g_t (1+g_t)^{\phi-1}, \tag{12}$$

$$\tilde{k}_{t+1} = \frac{1}{(1+n)(1+g_t)}(s\tilde{k}_t^{\alpha}(1-s_R)^{1-\alpha} + (1-\delta)\tilde{k}_t). \tag{13}$$

FIGURE 9.4 The independent dynamics of g_t



First, consider the independent dynamic equation (12) for g_t. You will easily verify that the transition curve defined by this equation has properties as illustrated in Fig. 9.4 (remember that $g_t > 0$, so Fig. 9.4 is only drawn for positive values of g_t and g_{t+1}). First, it passes through (0, 0). Second, for any g_t, its slope, $(1+n)^{\lambda}(1+g_t)^{\phi-2}(1+\phi g_t)$, is strictly positive.[3] Third, because we assume $n > 0$ there is a unique *positive* intersection with the 45° line: setting $g_{t+1} = g_t$ in (12) gives $(1+g_t)^{1-\phi} = (1+n)^{\lambda}$, or

$$g_t = (1+n)^{\frac{\lambda}{1-\phi}} - 1 \equiv g_{se}, \tag{15}$$

where indeed $g_{se} > 0$, because $n > 0$. Fourth, the slope of the transition curve at $g_t = 0$ is $(1+n)^{\lambda}$, and hence it is strictly larger than 1, because we assume $n > 0$.

The four features we have stated imply that g_t goes to g_{se} as t goes to infinity, as illustrated by the staircase-formed iterations in Fig. 9.4. This is one major conclusion from our model with $\phi < 1$ and $n > 0$: *in the long run the growth rate of knowledge converges monotonically to the particular value* $g_{se} > 0$. (You should note carefully how this conclusion depends on our assumption $n > 0$. An exercise will ask you to analyse the case $n < 0$ with respect to the evolution of g_t and the implied evolution of other endogenous variables.)

Next, consider the other dynamic equation, (13). As mentioned, this is not a real transition equation for $\tilde{k}_t$ because it has g_t on the right-hand side. However, we now know that in the long run g_t converges monotonically to the particular value g_{se}. We could therefore write down a new dynamic equation, setting the g_t in (13) constantly equal to g_{se}.

3. The slope of the transition curve (12) is found as follows:

$$\begin{aligned}\frac{dg_{t+1}}{dg_t} &= (1+n)^{\lambda}[(1+g_t)^{\phi-1} + (\phi-1)g_t(1+g_t)^{\phi-2}]\\ &= (1+n)^{\lambda}(1+g_t)^{\phi-2}[1+g_t+(\phi-1)g_t]\\ &= (1+n)^{\lambda}(1+g_t)^{\phi-2}(1+\phi g_t).\end{aligned}$$

If, according to this new equation, $\tilde{k}_t$ converges to a specific value $\tilde{k}^*$ in the long run, this will also be the implication of the correct dynamic system consisting of (12) and (13). The new equation would look almost exactly like our transition equations studied earlier, for instance the one in Chapter 5, and would, by similar reasoning, imply convergence of $\tilde{k}_t$ to a constant value $\tilde{k}^*$ (provided that $n + g_{se} + \delta + ng_{se} > 0$, which is fulfilled here because we assume $n > 0$).

Inserting g_{se} for g_t in (13) and solving for the constant solution, $\tilde{k}_{t+1} = \tilde{k}_t = \tilde{k}^*$, gives:

$$\tilde{k}^* = \left(\frac{s}{n + g_{se} + \delta + ng_{se}}\right)^{\frac{1}{1-\alpha}}(1 - s_R). \tag{16}$$

We can find the associated $\tilde{y}^*$ by using $\tilde{y}^* = (\tilde{k}^*)^{\alpha}(1 - s_R)^{1-\alpha}$:

$$\tilde{y}^* = \left(\frac{s}{n + g_{se} + \delta + ng_{se}}\right)^{\frac{\alpha}{1-\alpha}}(1 - s_R). \tag{17}$$

These expressions have well-known structures except for the $(1 - s_R)$ on the right-hand side, reflecting that now only this fraction of the labour input is used in the final goods sector.

We have shown that in the long run g_t converges to g_{se}, and $\tilde{k}_t$ and $\tilde{y}_t$ converge to $\tilde{k}^*$ and $\tilde{y}^*$, respectively. The situation where these variables are constant at their long-run values defines steady state.

We can illustrate the convergence process to steady state in two, one-way coupled diagrams. The first is Fig. 9.2 associated with the independent dynamic equation, (12), for g_t. The diagram shows how the technological growth rate, g_t, gradually converges to g_{se}. The other diagram is much like a modified Solow-diagram and it is associated with a rewriting of the other dynamic equation, (13), where one has the growth rate in $\tilde{k}_t$ on the left-hand side:

$$\frac{\tilde{k}_{t+1} - \tilde{k}_t}{\tilde{k}_t} = \frac{1}{(1+n)(1+g_t)}[s(1-s_R)^{1-\alpha}\tilde{k}_t^{\alpha-1} - (n + g_t + \delta + ng_t)]. \tag{18}$$

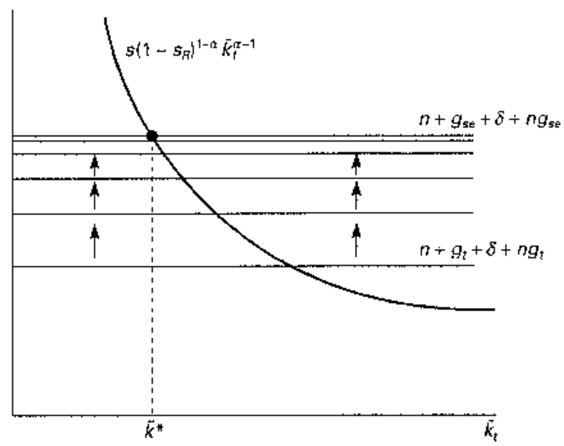
For any fixed g_t this is like a modified Solow equation and we illustrate it in Fig. 9.5.

For a given g_t in period t, the growth rate in $\tilde{k}_t$ will be proportional to the distance between the decreasing curve, $s(1 - s_R)^{1-\alpha}\tilde{k}_t^{\alpha-1}$, and the horizontal line at $n + g_t + \delta + ng_t$, and $\tilde{k}_t$ will move towards the intersection between the curve and the line. This time, however, the intersection is a 'moving target', since the horizontal line moves when g_t changes. In the long run the movements in g_t become smaller and smaller as g_t eventually converges to g_{se}, so the horizontal line gradually comes to rest at the position corresponding to g_{se}, as illustrated.

These diagrams provide intuition for an important insight. Assume that the economy starts in an 'old' steady state given by the (final) intersection points in Figs 9.4 and 9.5. Consider a parameter change that only affects the second of the diagrams, say a change in the depreciation rate δ. This will have no impact on Fig. 9.4, so the growth rate of A_t will stay unchanged at its old steady state value. The horizontal line in Fig. 9.5 will shift once and for all and thus initiate a process of dynamic adjustments which are like those in the Solow model (since g_t does not change, there will be no further changes of the horizontal line in Fig. 9.5). The speed of convergence will therefore be as in the Solow model.

Consider now a parameter change that shifts the transition curve in Fig. 9.4, say a change in λ. This will initiate adjustments in g_t, gradually taking it towards its new steady state value. Every time g_t changes, the horizontal line in Fig. 9.5 will shift. Convergence in Fig. 9.5 will now be towards a moving target and the convergence time will therefore be longer for

FIGURE 9.5 The 'modified Solow diagram'



this latter parameter change. In conclusion, a parameter change that initiates an adjustment process for g_t, that is, a change that somehow affects the research sector, will imply slower convergence to the new steady state than a parameter change that does not affect the research sector. This intuition is important for the theory we present in Section 9.7.

Growth in steady state

In steady state, when $\tilde{k}_t \equiv k_t/A_t$ and $\tilde{y}_t \equiv y_t/A_t$ have reached their long-run constant values, both k_t and y_t must be increasing at the same rate as A_t. In steady state this rate is g_{se}. Hence, in the long run, technology, A_t, capital per worker, k_t, and output per worker, y_t, converge towards growing at one and the same constant rate, g_{se}.

This is in accordance with the requirements for balanced growth. In particular, in steady state the capital–output ratio, $z_t \equiv K_t/Y_t = \tilde{k}^*/\tilde{y}^*$, will be constant and given by:

$$z^* = \frac{s}{n + g_{se} + \delta + ng_{se}}. \tag{19}$$

In steady state the rate of technological growth as well as the rate of economic growth is $g_{se} = (1+n)^{\lambda/(1-\phi)} - 1$. This is only positive if $n > 0$. Because of this feature our growth model with $\phi < 1$ is referred to as one of *semi-endogenous* growth (explaining the subscript *se* on g).

The intuition for population growth being a requirement for economic growth is this: when $\phi < 1$, a constant labour input into the research sector will imply that the growth rate, g_t, of technology will go to 0 in the long run as explained in the previous section. To sustain a positive constant growth rate in A_t, an increasing labour input into the research sector is required, and with a constant fraction, s_R, of all labour used in the research sector, this can only result if the labour force increases all the time.

Our formula, $g_{se} = (1+n)^{\lambda/(1-\phi)} - 1$, has the implication that $\lambda(1-\phi) \approx g_{se}/n$.[4] For magnitudes to be reasonable by Western standards, annual population growth rates in the range between $\frac{1}{2}$ and 1 per cent per year should be compatible with a steady state growth in

4. The formula for the growth rate may be written as $1 + g_{se} = (1+n)^{\lambda/(1-\phi)}$. Taking logs on both sides and using the approximation $\ln(1+x) \approx x$, we get $g_{se} \approx \lambda/(1-\phi)n$, or $g_{se}/n \approx \lambda/(1-\phi)$.

GDP per worker of around 2 per cent per year. Hence, $\lambda(1-\phi)$ should be in the range of 2–4. Assuming λ around 1 (not too much stepping on toes), ϕ should then be somewhere between $\frac{1}{4}$ and $\frac{1}{2}$.

As usual, we are more interested in output per worker, y_t, than in output per effective worker, $\tilde{y}_t$. From $y_t = \tilde{y}_t A_t$, output per worker must, in steady state, develop as $y_t^* = \tilde{y}^* A_t$. Furthermore, in steady state there is a necessary link between A_t and L_t. Note that (11) expresses the growth rate, g_t, as a function of A_t and L_t. In steady state $g_t = g_{se}$. In that case (11) reads: $\rho A_t^{\phi-1}(s_R L_t)^\lambda = g_{se}$, and therefore in steady state:

$$A_t = \left(\frac{\rho(s_R L_t)^\lambda}{g_{se}}\right)^{\frac{1}{1-\phi}}. \tag{20}$$

Using (17) and (20) plus the fact that $y_t^* = \tilde{y}^*/A_t$, it follows that steady state output per worker will evolve according to:

$$y_t^* = \left(\frac{s}{n + g_{se} + \delta + n g_{se}}\right)^{\frac{\alpha}{1-\alpha}} (1-s_R) s_R^{\frac{\lambda}{1-\phi}} \left(\frac{\rho}{g_{se}}\right)^{\frac{1}{1-\phi}} L_t^{\frac{\lambda}{1-\phi}}, \tag{21}$$

where we can insert that:

$$L_t^{\frac{\lambda}{1-\phi}} = [L_0(1+n)^t]^{\frac{\lambda}{1-\phi}} = L_0^{\frac{\lambda}{1-\phi}}(1+n)^{\frac{\lambda}{1-\phi}t} = L_0^{\frac{\lambda}{1-\phi}}(1+g_{se})^t, \tag{22}$$

thus tracing output per worker in steady state back to initial values and parameters (some of which are in g_{se}). The steady state growth path for consumption per worker may be found as $c_t^* = (1-s)y_t^*$.

Structural policy and empirics for steady state

Structural policy aimed at improving the steady state should be concerned with the steady state growth path of consumption per worker, its *level* as well as the *growth rate* along it.

With respect to the *level* this is influenced by the savings and investment rate, s, in the same way as in our models of exogenous growth with a golden rule value of s (the value that maximizes the height of the growth path) equal to capital's share, α (show this). There is also a golden rule value for the research share, s_R, and an exercise will ask you to determine this and explain the counteracting forces of a larger s_R on consumption per worker in steady state. A conclusion is that up to a relatively large value of the research share, a higher such share will elevate the steady state growth path of consumption per worker. Hence, policies to obtain reasonable large values of the investment rate and the research share seem well motivated by the concern to create a high level of consumption per worker in the long run. Note that the growth rate, n, of the labour force affects the level of the steady state growth paths of income and consumption per worker negatively. This is due to the well-known thinning-out-of-capital effect.

Taking s_R as given, the *growth rate* along the steady state growth paths, that is g_{se}, can only be increased by a policy that manages to increase the labour force growth rate, n (taking the technical parameters ϕ and λ as given also). As we discussed at length in Chapter 8, it is hard to reconcile empirical evidence with the feature that economic growth should depend positively on population growth.

According to our model with a constant labour force, a gradual and steady increase in the research share, s_R, will, as long as it can take place, have an influence on growth similar to that of an increasing labour force at a constant research share. In both cases labour input in the research sector will be increasing, which creates technological growth. A policy intended to increase the research share gradually thus seems to be a cautious alternative to one intended to promote population growth, since such a policy avoids the erosion of capital and natural resources per worker implied by growth in the labour force. A deeper discussion of *how* to influence the research share would require an endogenous determination of s_R and therefore (again) an explicit incorporation of the microeconomics of idea production, but obviously some policy instruments to consider are government subsidies to, or tax deductions for, private research activities as well as direct public involvement in research and education and public investment in universities and other research and educational institutions.

R&D-BASED SEMI-ENDOGENOUS GROWTH AND STRUCTURAL POLICY

Assuming a strictly positive growth rate of the labour force, and assuming that the elasticity of existing knowledge in the production of new knowledge is positive, but less than 1, the R&D-based growth model implies that the technological growth rate converges to a specific steady state value, g_{se}, and that capital, income and consumption per worker converge on steady state growth paths along which the growth rate is g_{se}, so that the capital–output ratio converges to a constant value in accordance with balanced growth. The growth rate, g_{se}, is strictly positive if and only if the population (labour force) growth rate is positive.

The **levels** of the long-run balanced growth paths depend on structural parameters in a way that supports policies to establish a high investment rate, a low population growth rate, *and* a high research share. The **growth rate** along the balanced growth paths depends on parameters in a way suggesting that a policy promoting population growth should also promote economic growth. Given that a positive association between population growth and economic growth does not seem very plausible empirically, a cautious alternative policy that would avoid a negative impact on the level of the growth path and would instead gradually raise this level could be a policy securing a gradual increase in the research share.

If, on empirical grounds, one has problems accepting that the source of long-term economic growth should be growth in the labour force, one should not necessarily conclude that our model with $\phi < 1$ is wrong, but only that *its steady state* does not seem to be descriptive. We may therefore pursue an idea similar to that used in Chapter 8. If convergence in our model is very slow, the empirical evidence may be in accordance with the model. A decrease in the population growth rate, n, will raise the steady state growth path as given by (21) and (22), but also reduce the growth rate along it. The first effect gives a transitory additional growth, while the latter effect gives a lower growth rate in the long run. If it takes a long time to come close to the new steady state growth path, the first effect will dominate for a long time. However, if convergence is so slow that the transitory growth effect dominates over time horizons up to 100 years, then we would not be so interested in the model's steady state. We would be more interested in the convergence process itself.

As in Chapter 8, this suggests studying the model for parameter values that imply very slow convergence. Again, a large ϕ (below 1) will have that effect. Just look at the transition equation, (12), for g_t. As ϕ goes to 1, the transition curve will come close to a straight line with slope $(1+n)^\lambda$ implying, for $n > 0$, a very long-lasting convergence of g_t to a very high value. At the limit, for $\phi = 1$, the technological growth rate, g_t, will, with $n > 0$, be exploding and go to infinity, which is, of course, absurd. We can eliminate the exploding g_t by considering $n = 0$. This is exactly what we will do next.

9.6 Endogenous growth

As an approximation of a large ϕ below 1, we will consider $\phi = 1$. To have a well-behaved model we will assume $n = 0$. We will denote the constant labour force by L.

It is easy to analyse the model that results from imposing these parameter restrictions on our base model consisting of Eqs (5)–(10). In fact, we have more or less analysed it before, since the model is almost identical to the Solow model considered in Chapter 5, with zero population growth and with a particular *endogenous* value for the growth rate of technology.

Equation (8), describing the evolution of the labour force, is now replaced by $L_t = L$ for all t. It follows from (9) and (10) that $L_{Yt} = (1 - s_R)L$ and $L_{At} = s_R L$ for all t. If we insert the latter of these into (6), we get

$$A_{t+1} - A_t = \rho A_t (s_R L)^{\lambda}, \tag{23}$$

or equivalently:

$$g_t \equiv \frac{A_{t+1} - A_t}{A_t} = \rho(s_R L)^{\lambda} \equiv g_e. \tag{24}$$

As stated earlier, when $\phi = 1$, a constant labour input $s_R L$ into the research sector gives a constant growth rate in technology, and this growth rate is now denoted by g_e (e for endogenous). Hence, $A_{t+1} = (1 + g_e)A_t$ for all periods t. Inserting $L_{Yt} = (1 - s_R)L$ in the production function for final goods, (5), and repeating the capital accumulation equation, (7), the full model reduces to the three equations

$$Y_t = K_t^{\alpha} (A_t (1 - s_R) L)^{1-\alpha},$$

$$K_{t+1} = sY_t + (1 - \delta)K_t,$$

$$A_{t+1} = (1 + g_e)A_t,$$

where one should bear in mind the endogenous value of g_e stated in (24). This model is like a Chapter 5, general Solow model with no population growth and an exogenous growth rate of technology g_e, with the one exception that only the fraction $(1 - s_R)$ of the constant labour force, L, is used in the production of final goods. With our usual definitions, $\tilde{y}_t \equiv Y_t/(A_t L_t) = y_t/A_t$, etc., it follows from an analysis similar to that in Chapter 5 (do this analysis) that $\tilde{y}_t$ converges to

$$\tilde{y}^* = \left(\frac{s}{g_e + \delta}\right)^{\frac{\alpha}{1-\alpha}} (1 - s_R),$$

and hence in the long run, y_t will converge towards the growth path:

$$y_t^* = \left(\frac{s}{g_e + \delta}\right)^{\frac{\alpha}{1-\alpha}} (1 - s_R) A_0 (1 + g_e)^t. \tag{25}$$

Consumption per worker will consequently converge towards the growth path $c_t^* = (1 - s)y_t^*$.

The model considered in this section has some obvious merits. First, it is a model of 'truly' endogenous growth. Without assuming exogenous technological growth, we get a constant and positive technology growth rate, g_e, and in the long run the growth rate of GDP per worker converges to g_e. Second, we have an expression showing how the growth rate depends on behavioural model parameters. Third, as can be seen from the expression given in (24), population growth is not a prerequisite for economic growth.

The main potential implication for structural policy is that in order to obtain a high growth rate in output and consumption per worker, the government should try to obtain a high *level* of the R&D share, s_R (whereas, under semi-endogenous growth, an *increasing* R&D share was needed to promote growth in the long run). This could, for instance, be done by subsidizing or undertaking research activities, as most governments indeed do.

The growth rate g_e given by (24) does not depend on the investment rate s, in contrast to what was found in the externality-based endogenous growth model of Chapter 8. This feature is an artefact, however, arising from the assumption of an exogenous research share. In a properly micro founded model, where s_R is endogenous, this share turns out to depend positively on the investment rate, s. The mechanism is basically that a higher savings rate will create more capital and therefore a lower real interest rate. This in turn means that the capital value of a patent based on a new idea and giving rise to a future stream of income will increase, thus making the private research sector more profitable. Hence, according to an appropriate interpretation of the model, policies to stimulate saving and investment are growth promoting.

Another merit of the model is that it does have convergence (unlike the externality-based endogenous growth model we studied in Chapter 8). Whereas the growth rate of technology is constant, the growth rate of output per worker varies over time as y_t gradually approaches its long-run growth path given by (25). In particular, for given values of all the parameters in (25), a lower initial GDP per worker, y_0, will imply a higher growth rate in GDP per worker. The model can, therefore, be considered to be in accordance with the observed conditional convergence between the countries in the real world. Still, since the model is explicitly intended to describe the world economy, it does not literally have much to say about cross-country evolutions.

The model also has some less attractive features. First, there is the objection that the model only pertains to a knife-edge case ($\phi = 1$). Just as we did not think that the knife-edge property was a serious drawback of the endogenous growth model of Chapter 8, we do not think it is a serious problem in the present model because we only consider $\phi = 1$ as an approximation of a large ϕ below 1.

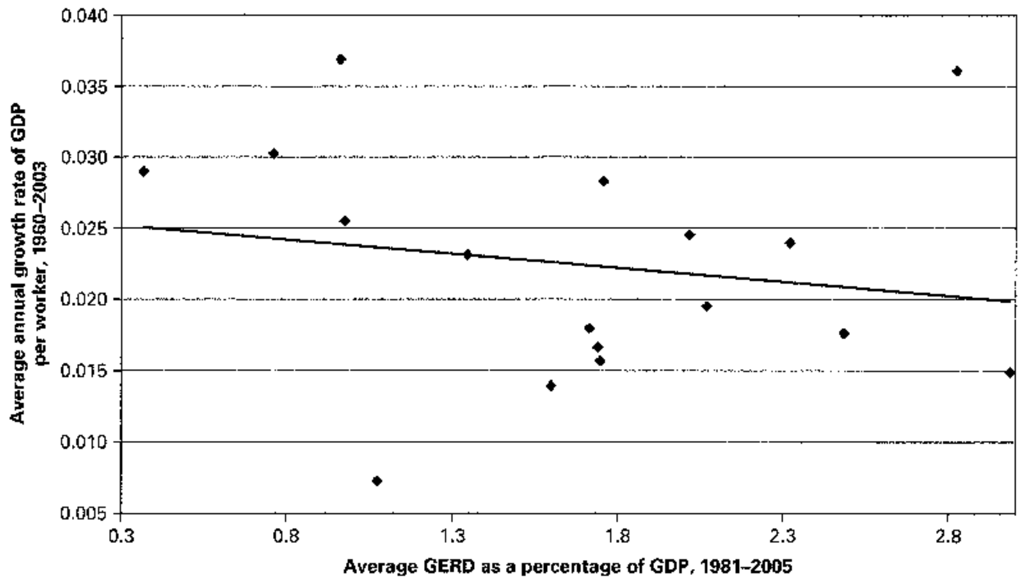
A main prediction of the model is that the long-run growth rate of GDP per worker and of technology should depend positively on the R&D share. The model is not intended to describe individual countries, but nevertheless it is tempting to look at cross-country evidence for growth rates and R&D shares. Figure 9.6 plots average annual growth rates in GDP per worker against GERD (recall this variable from Table 9.1) as a percentage of GDP across the 17 OECD countries of Table 9.1 for which we also have the data for GDP growth. Figure 9.7 plots average annual growth rates of A_t (using the same data as for the construction of Fig. 8.5) against GERD relative to GDP for the 14 countries for which we have the data needed.

Both figures suffer from having few observations and no clear relationship. They do not, however, point to a tight positive connection between growth rates and R&D shares across countries. As mentioned, the model does not describe individual countries, so cross-country evidence is, perhaps, not relevant.

The main problem of the model with $\phi = 1$ is the severe *scale effect* it implies. Equation (24) shows that a larger constant labour force, L, should, for a given s_R, give a higher growth rate in technology and then in the longer run also a higher growth rate in GDP per worker. If labour input in the research sector grows over time, either because L_t grows or because s_R grows or both, the result is a growing technological growth rate, g_t, as follows from (11) with $\phi = 1$. This will eventually lead to an increasing growth rate in output per worker. Empirical evidence suggests that indeed labour inputs into the research sectors in Western countries have been growing. Table 9.1 points to increasing R&D shares, and also independent longer-run evidence on the number of engineers and researchers employed in R&D activities, in particular in the USA, point to substantial increases. This should, according to the model with $\phi = 1$, have resulted in increasing growth rates in GDP per worker. However, this is not observed. The evidence points to constant long-term growth rates.[5]

5. The combination of gradually increasing labour input in the R&D sector and relatively constant long-run growth rates of GDP per worker is the opening observation of the article by Charles I. Jones mentioned in Footnote 1. Jones concludes that a value of ϕ equal, or close, to 1 is not realistic. For this reason he studies the model with ϕ (considerably) below 1, leading to semi-endogenous growth. Jones thinks that the very long-run evidence we discussed in Chapter 8 is supportive of the idea of semi-endogenous growth.

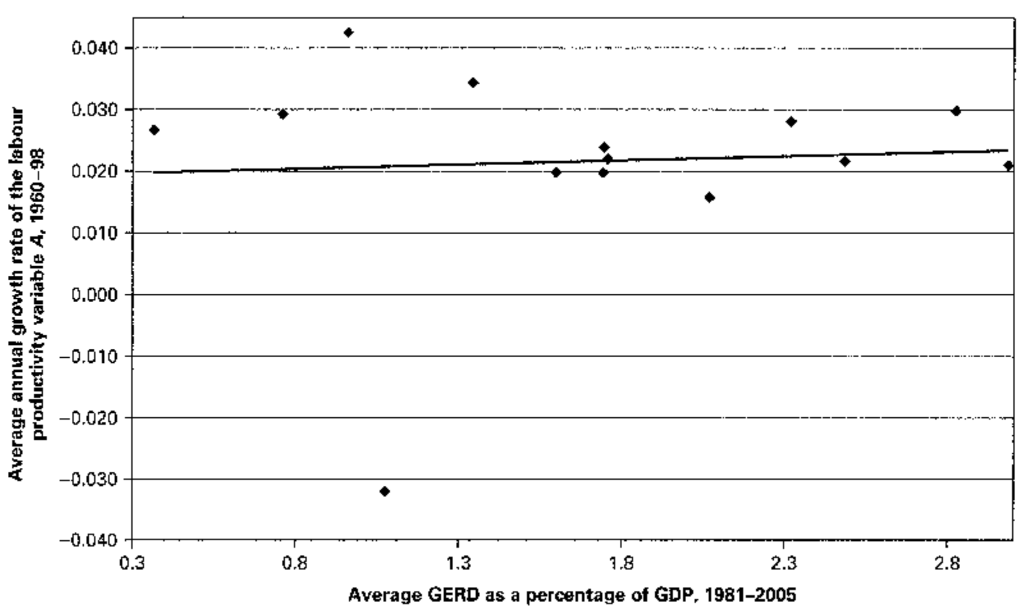
FIGURE 9.6 Average annual growth rate of real GDP per worker against GERD as a fraction of GDP, 17 OECD countries



Note: The measure of GERD is the average of GERD in 1981 and 2005.

Source: OECD Science & Technology Data Base, 2007, and Penn World Tables 6.2.

FIGURE 9.7 Average annual rate of labour-augmenting technological progress against average GERD as a fraction of GDP, 14 OECD countries



Note: The measure of GERD is the average of GERD in 1981 and 2005.

Source: OECD Science & Technology Data Base, 2007, and data set for Bernanke and Gürkaynak (2001).

R&D-BASED ENDOGENOUS GROWTH AND STRUCTURAL POLICY

The R&D-based growth model with an elasticity from existing knowledge to the production of new knowledge in the R&D sector's production function equal to one implies a constant technological growth rate, g_e, even with a constant research share and no growth in the labour force. The endogenous growth rate g_e depends positively on the research share and (under an additional plausible mechanism through which the research share depends positively on the savings rate) on the savings rate. Capital, income, and consumption per worker all converge gradually towards steady state growth paths along which the growth rate is g_e. Policies that increase the research share and the savings rate are thus growth-promoting policies. The model is, however, plagued by an implausible scale effect according to which a larger labour input in the R&D sector should result in a permanently larger growth rate of GDP per worker.

9.7 'Hemi-endogenous' growth[6]

The model we consider in this chapter paves the way for a third interpretation of the past 200 years of economic growth in the developed world, viewing this as neither semi-endogenous nor truly endogenous growth. We will use the term 'hemi-endogenous' growth for this third interpretation, because it may be seen as an attempt to explain the process of economic growth in the Western hemisphere.

It is (plausibly) assumed that $\phi < 1$, so an increasing labour input in the research sector is required to sustain economic growth for long periods. According to the hemi-endogenous growth interpretation, this increase in labour input does not primarily result from growth in the labour force, but mostly from steady ongoing growth in the research share. The model considered is like the model of semi-endogenous growth with $\phi < 1$, so it does have a steady state with a positive relationship between population growth and economic growth. However, the view taken here is that the real world is always on its way to steady state in response to new increases in the research share. Of course, steady growth in the research share cannot go on forever, as this share has a maximum of 1, but growth at a small positive rate can take place for a long time, particularly if the share is initially small.

The possibility of the hemi-endogenous growth interpretation is linked to a certain property of the model we have studied in this chapter: even if no parameter is initially at an extreme value (such as $\phi = 1$), certain parameter changes can have very long-lasting effects. We have already given the intuition for this in Section 9.3. If a parameter change does *not* affect the (independent) dynamics of the technological growth rate, g_t, the response to it will be as in the Solow model, also with respect to its duration. If a parameter change *does* affect the dynamics of g_t, the economy's response to it will potentially be longer-lasting since it implies all the types of adjustments found in the general Solow model *plus* the adjustments of g_t. The latter kind of adjustments give the first ones the character of 'convergence towards a moving target'.

To illustrate, we will conduct simulations with our model showing the economy's responses to a change in the investment rate, s, and to a change in the R&D share, s_R. The first of these changes does not affect the research sector and hence does not initiate an adjustment process for g_t, while the other does affect the research sector and therefore triggers adjustments in g_t.

Let us specify non-extreme initial parameter values that are reasonable for a period length of one year. First we set $\alpha = \frac{1}{3}$, $\delta = 0.06$, and $s = 0.2$, which are normal values. From

6. This section discusses an alternative interpretation of growth in the Western world during the past 200 years. The interpretation is related to the one suggested in Charles I. Jones, 'Sources of U.S. Economic Growth in a World of Ideas', *American Economic Review*, **92**, 2002, pp. 220–239. Reading the section is hopefully rewarding, but not required for further reading of the book.

Table 9.1 it could be reasonable to let $s_R = 0.02$. An annual population growth rate of 1 per cent also seems reasonable in a 200-year perspective, so we set $n = 0.01$, but note that assuming $n = 0.005$ would not change our conclusions substantially. For the parameters relating to the research sector we set $\rho = 1$ as a pure normalization, and we choose $\lambda = 1$ and $\phi = \frac{1}{2}$. This implies that the steady state annual growth rate, $g_{se} = (1+n)^{\lambda(1-\phi)} - 1$, is around 0.02, which seems to be a reasonable anchor for our simulations.

Let us also say that in year 0 the economy is in steady state at these parameter values, and set the initial value of the state variable L_t at $L_0 = 1$. The initial values, K_0 and A_0, of the other state variables follow from the requirement that the economy is in steady state. First, A_0 should obey the necessary steady state link between A_0 and L_0 given in (20), here implying $A_0 \approx 0.99$. Finally, the capital–output ratio in steady state is given by (19) which imposes a restriction, $K_0/Y_0 = z^*$, on the initial capital–output ratio, and K_0 and Y_0 must also fit in the production function, $Y_0 = K_0^{\alpha}(A_0(1-s_R)L_0)^{1-\alpha}$. With our assumed parameter values and the already determined values for L_0 and A_0, solving the two equations in K_0 and Y_0 gives $K_0 \approx 3.20$ (verify this).

We now have values for all parameters and initial values for the state variables. We can then simulate the model consisting of Eqs (5)–(10) over a number of years. We will do so for 200 years, assuming a permanent parameter change in period 10. We first consider an increase in the investment rate from the initial value $s = 0.2$ up to $s' = 0.3$.

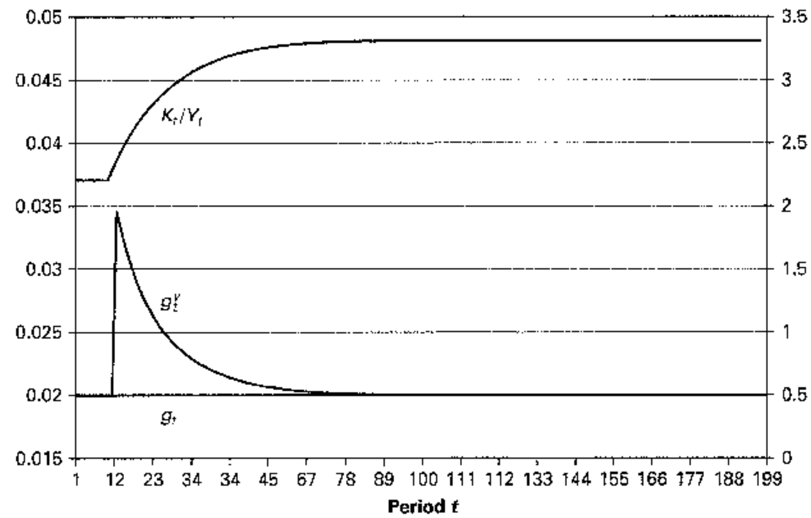
Before showing the results of the simulations, let us think about the qualitative effects using our diagrams, Figs 9.4 and 9.5 (do the drawing yourself). Since the dynamics for g_t are unaffected by s, as shown by the dynamic equation (12), the first of the diagrams is unaffected. Since g_t is initially at its steady state value g_{se}, it will stay at that value during all 200 years. In the second diagram the shift in the investment rate implies that the decreasing curve shifts upwards once and for all, which initiates a process of gradual increases in $\tilde{k}_t$, the growth rate of $\tilde{k}_t$ being largest at the beginning and then falling back towards 0 as $\tilde{k}_t$ approaches its new steady state value. Consequently, since $\tilde{k}_t = k_t/A_t$, the growth rate of k_t will jump up from g_{se} and then gradually return to g_{se}. The same will be true for the growth rate of y_t.

Figure 9.8 shows some output from the simulation and confirms the qualitative findings. It also gives an impression of the duration of the adjustments. Note that g_t stays at the unchanged steady state value g_{se} just above 0.02 all the time, while g_t^y (the approximate growth rate of y_t) jumps to around 0.035 and then starts to fall towards g_{se}. After 15 years it passes through the value 0.025, so at this point one half of a percentage point of the additional growth remains.

Now turn to a permanent change in the R&D share, assuming that it increases from the value $s_R = 0.02$ to $s_R' = 0.03$ in year 10, a 50 per cent increase as with the savings rate. How does this affect a diagram like Fig. 9.2 (you should draw the diagram yourself)? Looking at (12), you may first think that it does not affect it at all, since s_R does not enter into (12). However, in the derivation of (12) we assumed that s_R was constant, but now there is a change in this parameter in one period. In this one period, (12) does not hold. In the first period where the new s_R' works, that is, in period 10, the technological growth rate (up to the next period) as given by (11) is $g_{10} \equiv (A_{11} - A_{10})/A_{10} = \rho A_{10}^{\phi-1}(s_R' L_{10})^{\lambda}$, while in period 9, $g_9 = \rho A_9^{\phi-1}(s_R L_9)^{\lambda} = g_{se}$. Dividing the first by the second, etc., gives: $g_{10} = g_9(A_{10}/A_9)^{\phi-1}(s_R'/s_R)^{\lambda}(L_{10}/L_9)^{\lambda} = g_9(1+g_9)^{\phi-1}(s_R'/s_R)^{\lambda}(1+n)^{\lambda}$. Here, since in period 9 we are in steady state, it follows from (12) that $(1+n)^{\lambda}g_9(1+g_9)^{\phi-1} = g_{se}$. Inserting this gives $g_{10} = (s_R'/s_R)^{\lambda} g_{se}$. That is, in period 10 the technological growth rate, g_t, jumps by the factor $(s_R'/s_R)^{\lambda}$. After that, since the R&D share is again constant, the transition curve in Fig. 9.4 is unchanged and the associated dynamics gradually take g_t back towards the steady state value g_{se} around 0.02. In Fig. 9.5 this means that the horizontal line first jumps up and then gradually falls down towards its original position. Hence, eventually the growth rates of both A_t and y_t return to g_{se}.

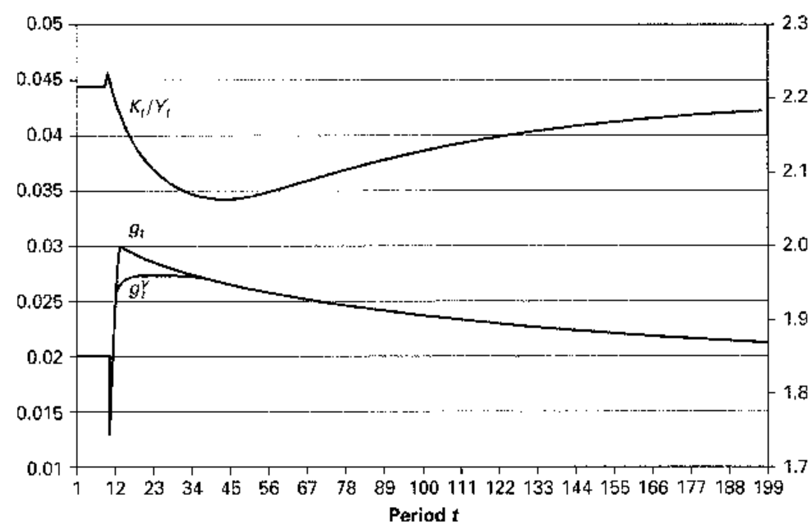
Figure 9.9 reports on the simulation and gives an impression of the duration of the adjustments. Note how g_t jumps up once and then falls gradually back towards 0.02. The growth rate of final output per worker initially falls, reflecting that a first effect of the

FIGURE 9.8 Simulation, evolution of g_t, g_t^y, and K_t/Y_t after an increase in s



Note: Growth rates left scale. Capital share of output right scale.

FIGURE 9.9 Simulation, evolution of g_t^y, and K_t/Y_t after an increase in s_R



Note: Growth rates left scale. Capital share of output right scale.

increased s_R is that relatively fewer people will be working in the final goods sector. After that the higher growth in A_t will pull up the growth in y_t. The phase where A_t grows faster than y_t corresponds to the phase where $\tilde{k}_t$ is decreasing in Fig. 9.5, and the later phase where y_t grows faster than A_t is when $\tilde{k}_t$ begins to increase again.

The main conclusion from Fig. 9.9 is that the adjustments following the increase in the research share are long lasting. For instance, although the jump upwards in g_t^y in the beginning is not as large as after the increase in s, it takes almost 50 years before the growth rate of y_t falls below 2.5 per cent. Of course, the duration of the adjustments depends on the parameter values, but these were deliberately chosen to avoid extreme cases.

If a once and for all change in the R&D share has long-lasting effects on growth, then perhaps a slow and gradual increase in s_R, which is not unrealistic by empirical standards, can sustain growth at reasonable levels for a long time? This is exactly the possibility of hemi-endogenous growth that we now turn to.

Let us first assume that $n = 0$, so that population growth is not the factor that keeps economic growth alive. Let us also assume that the R&D share increases at a pace earlier suggested as reasonable. It grows by a constant annual rate, g_{s_R}, implying a doubling every 100 years: $(1+g_{s_R})^{100} = 2$, or $g_{s_R} = 2^{1/100} - 1 \cong 0.7$ per cent. As stated earlier, this cannot go on forever, but it could take place for 200 years, for instance by s_R increasing from 0.75 to 1.5 per cent over the first century and from 1.5 to 3 per cent over the next one. Consequently, the exogenous R&D share is no longer a constant, but an increasing sequence. How much growth in A_t can the assumed growth in s_R in itself sustain? Return to (11), write it down (once again) for period $t + 1$ as well as for period t, and divide the first equation by the second, this time remembering that s_R depends on time, while L_t is constant. You will easily verify that this gives:

$$g_{t+1} = (1+g_{s_R})^{\lambda} g_t (1+g_t)^{\phi-1}.$$

The *constant* growth rate that can be sustained (inserting $g_{t+1} = g_t$) is thus:

$$g = (1+g_{s_R})^{\lambda/(1-\phi)} - 1.$$

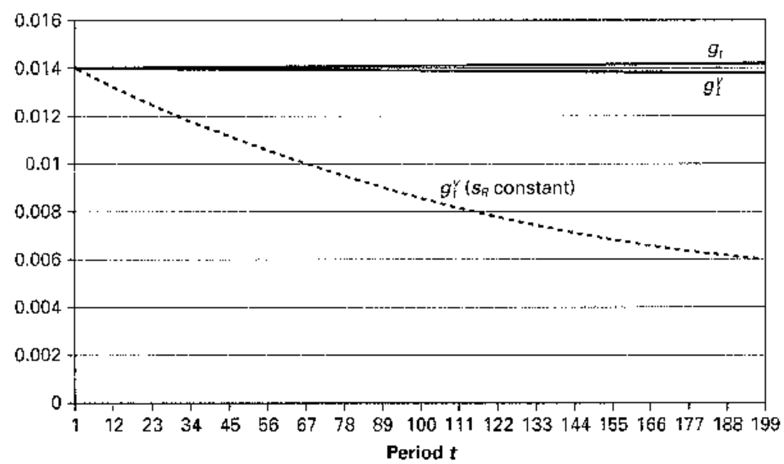
For the g_{s_R} assumed above, and $\lambda/(1-\phi) = 2$, this is $g = 2^{2/100} - 1 \approx 1.4$ per cent. Interestingly, without any population growth at all, a plausible-looking gradual increase in the R&D share can sustain continuing growth in the technological parameter, A_t, that is not far from the observed relatively constant growth rates in GDP per capita over long periods.

Let us assume that in year zero s_R indeed has the value 0.0075, and then grows for 200 years at an annual rate of $g_{s_R} = 2^{1/100} - 1$. As mentioned, we consider $n = 0$ and, as a pure normalization, we fix the constant labour force at $L = 1$. For all other parameters we assume the same values as the initial ones above, $s = 0.2$, $\rho = \lambda = 1$, $\phi = \frac{1}{2}$, etc. Let us also assume that the initial value, A_0, of the technological parameter is such that already from year 0, g_t takes the constant value g. From (11) this requires, $g = A_0^{\phi-1} s_{R0}$, or $2^{2/100} - 1 = 0.0075 A_0^{-1/2}$, which gives $A_0 \approx 0.29$. This means that A_t will be growing at an annual rate of around 1.4 per cent over all 200 years. Finally we calibrate the initial value K_0 of capital so that the growth rate of output per worker also starts at around 1.4 per cent. We then simulate the model over 200 years. Figure 9.10 shows the results. The annual growth rate of output per worker stays very close to 1.4 per cent over the entire 200 years. The lower curve shows how the growth rate of output per worker would evolve if s_R stayed constant at the initial 0.75 per cent.

The assumed plausible-looking gradual increase in the research share can thus, over time spans like 200 years, 'explain' economic growth rates close to the observed ones. If we had assumed a slightly faster increase in s_R, we could also have rationalized technological and economic growth rates of 1.8–2 per cent per year. Hence, one possible interpretation of the growth in the developed world over the past 200 years is that it is really a transitory growth that is being maintained all the time by gradual increases in the research *share*.

This is neither semi-endogenous nor endogenous growth. Qualitatively hemi-endogenous growth is not so different from the growth one could have in a basic Solow model if the investment rate were increasing gradually. The difference is that the research share is initially so small, and the impact of increases potentially so strong and long lasting, that the research share can keep on growing at a small constant rate for a very long time, and this suffices for creating economic growth rates of realistic magnitudes for long times. It is this kind of interpretation of the growth process we refer to as hemi-endogenous growth.

FIGURE 9.10 Simulation of the possible evolution of g_t^A and g_t^Y associated with a gradual increase in s_R



One fascinating consequence of the hemi-endogenous growth interpretation is that *it cannot go on forever.* We may expect a few more centuries of growth in GDP per person based on a doubling of the research share every 100 years, but s_R cannot keep on doubling forever. According to the hemi-endogenous growth interpretation, some day growth will cease. At that time, however, the constant income per capita may be very high.

THE HEMI-ENDOGENOUS GROWTH INTERPRETATION ACCORDING TO THE R&D-BASED GROWTH MODEL

A possible and fascinating explanation – or interpretation – of long-run growth experiences like those of the developed world over the past 200 years is that the constant growth in GDP per capita of around 1.5–2 per cent per year is really due to a slow and gradual increase in the amount of resources used in R&D activities corresponding to roughly a doubling of the 'research share' every one hundred years. If this is the right explanation, one day economic growth will be over. At that time, however, mankind could be rich.

9.8 Summary

1. This chapter has studied the microeconomics of private idea (R&D) production. The fact that ideas are non-rival implies that to induce private R&D, market power is needed in the selling of ideas and idea-based products. Otherwise the fixed development cost cannot be covered. The fact that ideas are only imperfectly excludable necessitates some arrangements to increase excludability and prevent copying in order for private production to take place.
2. Patent laws as well as other legal rules for the protection of intellectual property rights can serve the double purpose of ensuring both excludability and monopoly in the production and selling of idea-based products. This way of obtaining private R&D activities is, however, imperfect. Ensuring monopoly power to the sellers of ideas implies that ideas are priced above marginal cost and therefore, from a social viewpoint, underutilized.

3. Even though a patent system is not a perfect way of ensuring private research and development, it is effective in the sense that much more private R&D will take place in the presence than in the absence of laws that protect intellectual property rights. According to some economic historians, the unprecedented economic growth during the past 200 years is partly due to the fact that the legal protection of intellectual property rights reached a critical level some two centuries ago.
4. The macro model of R&D-based endogenous growth studied in this chapter abstracted from the subtle microeconomics of idea production and had the following features:
   - Final goods are produced from the inputs of capital, labour and (labour-augmenting) technology, A_t, according to a standard Cobb–Douglas production function.
   - New technology is produced in the R&D sector from the inputs of labour and existing technology according to a production function with constant returns to labour at the firm level but possibly diminishing returns at the aggregate level, because firms in the R&D sector may be 'stepping on each other's toes' by working on the same ideas. There is a specific elasticity, ϕ, of the output of new technology, $A_{t+1} - A_t$, with respect to the input of existing technology, A_t.
   - The labour force can be used in either production of final goods or in production of new technology, whereas existing technology can be used in both sectors, reflecting that ideas are non-rival. The fraction of labour used in the R&D sector is an exogenous parameter called the research (or R&D) share.
   - In other respects the model is standard: capital accumulates from savings and investment (minus depreciation), and the labour force grows at an exogenous rate.
5. If ϕ is equal to 1, a constant input of labour in the R&D sector creates a constant and positive growth rate of technology, $(A_{t+1} - A_t)/A_t$. If ϕ is less than 1, reflecting that it becomes harder to generate new ideas as the more obvious ideas have already been discovered, a constant labour input in research will imply that the absolute production of technology, $A_{t+1} - A_t$, increases and goes to infinity over time, but the growth rate, $(A_{t+1} - A_t)/A_t$, goes to 0. To maintain a positive, constant growth rate of technology requires a positive, constant growth rate of the labour input into the research sector (when $\phi < 1$).
6. In the case $0 < \phi < 1$, the model implies convergence to a steady state where capital and output per worker both grow at the same constant rate as technology, and the growth rate of technology depends positively on the rate of growth of the labour force (or population) in such a way that there can only be economic growth if there is population growth. The reason is that given a constant research share, the labour input in the research sector can only grow at a constant rate if population grows at a constant rate. The semi-endogenous growth arising in this case suggests that a policy to promote economic growth should be one that stimulates population growth.
7. In the alternative case where $\phi = 1$, a constant labour force and a constant research share imply a constant positive growth rate of technology that depends positively, and in a simple way, on the size of the research share. The full model gives convergence to a steady state where capital and income per worker both grow at the same rate as technology. A growth-promoting structural policy is one that increases the research share, but the model is tacit about how this can be achieved since the research share is exogenous. The model of truly endogenous growth arising when $\phi = 1$ implies a highly implausible scale effect: a larger labour force should imply a higher rate of growth of GDP per worker, and if the labour force increases at a constant rate, the growth rate of income per worker should explode.

8. Mainly because of empirical evidence concerning the links between population growth and economic growth, it can be hard to believe that the observed long-lasting economic growth in Western countries should be understood as either purely semi-endogenous growth (where growth in the labour force is needed for economic growth because the research share is constant, and $\phi < 1$) or as truly endogenous growth (where economic growth can occur for a constant labour force because $\phi = 1$, but an increasing labour force would imply increasing economic growth).
9. According to the model studied in this chapter there could be an in-between explanation of economic growth during the past 200 years. Hemi-endogenous growth, as we called it, is closely related to semi-endogenous growth (assuming $\phi < 1$), but explains the required growth of the labour input in R&D as the result, not of population growth given the research share, but of a gradually increasing research share possibly for a constant labour force. We showed that under reasonable parameter specifications, a realistic slow and gradual increase in the research share could, according to our model, explain annual growth rates of GDP per capita in the range 1.5–2 per cent over periods of 200 years and more. This makes the hemi-endogenous growth interpretation look plausible. Of course, the research share cannot grow at a constant rate forever, so one day hemi-endogenous growth will be over. When the research share reaches its maximum, increased labour input in the R&D sector (required for economic growth when $\phi < 1$) can only be achieved by increases in the labour force.

9.9 Exercises

Exercise 1. Rivalry and excludability

The figure below is taken with modifications from an article by Paul Romer.[7] It divides the plane of commodity characteristics according to two criteria: degree of excludability and degree of rivalry. While rivalry is a zero–one concept, excludability occurs more continuously in various degrees, in the figure reflected by three degrees. We have inserted a few commodities in the figure, according to their characteristics.

DEGREE OF EXCLUDABILITY ↑ / DEGREE OF RIVALRY ←	RIVAL	NON-RIVAL
FULLY EXCLUDABLE (OR CLOSE)	Potatoes, bread, beers	Passage of a bridge
INTERMEDIATELY EXCLUDABLE	Fish in the sea? (think of quotas, etc.)	Computer code for spreadsheet
NON-EXCLUDABLE (OR CLOSE)	Fish in the sea	Fresh air, national defence

7. Paul M. Romer, 'Two Strategies for Economic Development: Using Ideas and Producing Ideas', in Proceedings of the World Bank Annual Conference on Development Economics 1992, supplement to the *World Bank Economic Review*, March 1993, pp. 63–91.

Your task is to discuss and suggest an appropriate placement of the following commodities:

- Drinking a beer in a bar
- Enjoying the atmosphere in a bar
- Bohr's atomic theory
- Travels
- Great scenic views
- The concept for a TV show
- TV transmissions 'in the air' (discuss encoded versus non-encoded and the extent to which excludability of non-encoded signals can be obtained by legal arrangements)
- Biscuits
- The idea of the assembly line
- Mozart's *Eine Kleine Nachtmusik*
- The services of a dentist
- Basic R&D, like discoveries in basic research
- Cable TV signals
- The idea for a new type of motor for cars (discuss legal and technical ways to ensure more excludability)
- Taxi rides
- Parking in a car park at a shopping centre (discuss the implication of the size of the cost of enforcing a payment system)
- Pythagoras' theorem.

Now use your own imagination and place other examples of commodities in the figure, at least one for each of the cells.

Exercise 2. The R&D-based macro model of endogenous growth in continuous time

In continuous time the model we have studied in this chapter would be:

$$Y = K^{\alpha}(AL_Y)^{1-\alpha},$$

$$\dot{A} = \rho A^{\phi} L_A^{\lambda},$$

$$\dot{K} = sY - \delta K,$$

$$\frac{\dot{L}}{L} = n,$$

$$L_Y + L_A = L$$

$$L_A = s_R L.$$

Both notation and parameter restrictions are obvious and standard. We assume $\phi > 0$ and $\lambda \leq 1$.

1. Show that, at any time, the technological growth rate, $g_A \equiv \dot{A}/A$, according to this model is:

$$g_A = \rho A^{\phi-1} L_A^{\lambda} = \rho A^{\phi-1}(s_R L)^{\lambda}.$$

Explain for each of the two cases $\phi = 1$ and $\phi < 1$ how the output of the research sector $\dot{A}$, the technological level A, and the technological growth rate g_A evolve over time when the labour input into the research sector $L_A = s_R L$ is constant.

2. Still using standard notation and definitions, e.g. $\tilde{k} \equiv K/(AL) = k/A$ and $\tilde{y} \equiv Y/(AL) = y/A$, show that the model implies the two one-way coupled differential equations in g_A and $\tilde{k}$:

$$\dot{g}_A = (\phi - 1)g_A^2 + \lambda n g_A,$$

$$\dot{\tilde{k}} = s\tilde{k}^{\alpha}(1 - s_R)^{1-\alpha} - (n + g_A + \delta)\tilde{k}.$$

(Hint: for the first one take logs on both sides and then differentiate with respect to time in the above formula for g_A.)

Assume first that $\phi < 1$ and $n > 0$.

3. Illustrate the first of the above differential equations in a diagram with g_A along the horizontal axis and $\dot{g}_A$ along the vertical one. Show that g_A converges monotonically to:

$$g_{se} \equiv \frac{\lambda n}{1 - \phi}.$$

4. Illustrate the second equation in a 'Solow-type diagram' (and, if you wish to, also in a 'modified Solow-type diagram') and show that $\tilde{k}$ and $\tilde{y}$ converge to specific constant values, $\tilde{k}^*$ and $\tilde{y}^*$, respectively, in the long run. Show that the growth rates of capital per worker and of output per worker both converge to g_{se} in the long run.

Now consider the case $\phi = 1$ and $n = 0$, and let the constant labour force be denoted by $\bar{L}$.

5. Show that the technological growth rate is constantly $g_e = \rho(s_R\bar{L})^{\lambda}$ and that the growth rates of capital per worker and of output per worker both converge to g_e.

Exercise 3. Golden rule in the model of semi-endogenous growth

Consider the model studied in Section 9.5 where $0 < \phi < 1$ and $n > 0$, and the steady state growth paths for output per worker, y_t, (described in Eqs (21) and (22)) and for consumption per worker, $c_t = (1 - s)y_t$. What are the (golden rule) values of the fundamental rates, s and s_R, that will elevate the steady state growth path for consumption per worker to the highest position? Describe in words the two opposite effects of the R&D share, s_R, on the levels of the steady state growth paths. Say that reasonable values of λ and ϕ with a population growth rate of 1 per cent per year should imply a steady state growth rate g_{se} of 2 per cent per year. Show that this would mean that $\lambda(1 - \phi)$ should be around 2. What is the golden rule value for the R&D share under this assumption and how does it relate to the empirical GERD shares reported in Table 9.1?

Exercise 4. A scale effect also in the model of semi-endogenous growth

The model of endogenous growth studied in Section 9.6 was criticized for the scale effect it implied: an increasing labour input into the research sector, arising, for example, from population growth ($n > 0$), would imply ever-increasing technological and economic growth rates over time, which is empirically implausible. Consider now the model of semi-endogenous growth presented in Section 9.5. In this model, $n > 0$ does not imply increasing (but rather positive and constant) growth rates in the long run. Explain for yourself the basic intuition for this. The model can, nevertheless, be said to involve a 'scale effect', not on growth rates, but on levels. Explain this effect by stating how a larger level of the labour force, as measured by L_0, affects the level of the steady state growth path for output per worker given by (21) and (22). Do you find that this scale effect is as implausible as the one connected to the model of endogenous growth?

Exercise 5. The effects of an increase in research productivity

This exercise asks you to analyse qualitatively the effects of a once and for all, permanent increase in the productivity parameter ρ of the research sector. Consider the model of semi-endogenous growth ($0 < \phi < 1$ and $n > 0$). Assume that in all periods up to and including period 0 the economy has been in steady state at 'old' parameter values, among them $\rho > 0$. In particular, the growth rates $g_t \equiv (A_{t+1} - A_t)/A_t$ and $g_t^y \equiv (y_{t+1} - y_t)/y_t$ have been equal to $g_{se} = (1+n)^{\lambda/(1-\phi)} - 1$ in all periods up to and including period 0. Working first time in period 1 the productivity parameter jumps permanently to $\rho' > \rho$.

1. First show that this implies the technological growth rate from period 1 to 2: $g_1 = (A_2 - A_1)/A_1 = (\rho'/\rho)g_{se}$. (Hint: Write down Eq. (11) of Section 9.4 for period 1 and for period 0, divide the first by the second and proceed as in Section 9.4, remembering that $g_0^A = g_{se}$.)
2. Using a diagram like Fig. 9.4 for the independent dynamics of g_t, describe how g_t evolves over time. (Note that the transition curve does not move, but there is the initial jump in g_t you found in Question 1.)
3. Does the long-run growth rate of output per worker change as a result of the change in the research productivity parameter? What are the long-run effects on output and consumption per worker of the change?
4. In the model of endogenous growth ($\phi = 1$, $n = 0$), what is the long-run effect on the growth rate of GDP per worker of a permanent increase in ρ?

Exercise 6. Some special cases

Consider the transition equation (12) for g_t.

1. If $\phi = 1$ and $n > 0$, what does the transition curve look like and what is the evolution of the technological growth rate g_t in the long run? (Once again this is really just a description of the scale effect in the model of endogenous growth.)
2. Same questions for $\phi > 1$ and $n = 0$.
3. Same questions for $\phi < 1$ and $n < 0$, and for $\phi = 1$ and $n < 0$. For these ones, what is the long-run growth rate in output per worker according to the full model?

Exercise 7. The rate of convergence of the technological growth rate

Consider the transition equation (12) for g_t repeated here for convenience:

$$g_{t+1} = (1+n)^{\lambda} g_t (1+g_t)^{\phi-1} \equiv G(g_t).$$

Assume $\phi < 1$ and $n > 0$. Hence, in the long run g_t converges to $g_{se} = (1+n)^{\lambda/(1-\phi)} - 1$. In this exercise we are interested in the speed of convergence.

1. Sketch the transition curve in a diagram and explain how it shifts as ϕ increases and eventually goes to 1. How will this affect the speed of convergence?
2. Consider the generic form $g_{t+1} = G(g_t)$ of the transition equation. Show that linearizing this around the steady state value g_{se} and changing into logs yields:

$$\ln g_{t+1} - \ln g_t \approx (1 - G'(g_{se}))(\ln g_{se} - \ln g_t).$$

Argue that $(1 - G'(g_{se}))$ can be viewed as the rate of convergence for the technological growth rate g_t *according to the linear approximation* of the transition equation for g_t. Show that:

$$1 - G'(g_{se}) = (1-\phi)\left(1 - \frac{1}{(1+n)^{\frac{\lambda}{1-\phi}}}\right),$$

and that the rate of convergence goes to 0 as $\phi \to 1$.

3. Compute the range that the annual rate of convergence, $1 - G'(g_{se})$, will be in for values of λ between $\frac{1}{2}$ and 1, values of ϕ between $\frac{1}{4}$ and $\frac{3}{4}$, and values of n between 0.5% and 1% per year. Comment on the order of magnitude of $1 - G'(g_{se})$ for these non-extreme parameter

values, e.g., by comparing to the rate of convergence we typically find for output per effective worker in the Solow model. Figure 9.8 shows a relatively slow convergence of g_t for the particular parameter values $\lambda = 1$, $\phi = \frac{1}{2}$, and $n = 0.005$. Does this relatively slow convergence seem to be robust for parameters in the non-extreme region considered here?

Exercise 8. The effects of a decrease in the growth rate of the labour force

Consider a decrease in the growth rate n of the labour force in the model of semi-endogenous growth. We know from the chapter's analysis of the model that in the long run a lower n has a negative impact on economic growth, but due to transitory growth effects it may have a positive influence for some time. For some points in the chapter the duration of this positive effect can be considered important. If the positive impact can last for a long time (for reasonable parameter values), this may be what makes the model compatible with the empirical observation (Table 8.1) that over quite long periods lower population growth rates seem to go hand in hand with higher growth rates in GDP per capita within countries in the developed world.

Therefore, consider the model of semi-endogenous growth ($0 < \phi < 1$ and $n > 0$). Assume that in all periods up to and including an 'initial' one the economy has been in steady state at 'old' parameter values, among them $n > 0$. Working for the first time between the initial period and the next one, the growth rate of the labour force falls to a lower value $n' < n$, but still $n' > 0$.

1. To describe the qualitative effects, explain how diagrams like Figs 9.4 and 9.5 are affected by the parameter change and explain how g_t and $\tilde{k}_t$ adjust over time. Explain how there can be a positive contribution to growth in y_t in the beginning and why, in the long run, the growth rate of y_t has to fall.

   Assume parameter values, $\alpha = \frac{1}{3}$, $\rho = \lambda = 1$, $\phi = \frac{1}{2}$, $s = 0.2$, $\delta = 0.06$, $s_R = 0.03$ and initially $n = 0.015$. Starting between period 10 and 11, n drops permanently to $n' = 0.005$ while the other parameters stay unchanged.

2. Compute the steady state values for g_t, g_t^Y and K_t/Y_t before and after the parameter change. Simulate the model consisting of Eqs (5)–(10) over periods 0 to 200 assuming that from period 0–10 the economy is in steady state at the old parameter values. (As we did in Section 9.7, normalize, e.g., $L_0 = 1$, and find the initial values, A_0 and K_0, implied by the economy being in steady state in period zero.) Create figures like in Section 9.7 showing the evolution of g_t, g_t^Y and K_t/Y_t. Comment with respect to how long lasting the positive effect on g_t^y is.
3. Assume that the decrease in the growth rate of the labour force comes gradually, so the change from 0.015 to 0.005 takes place over the 120 years following period 10 with an equal relative change each year. Simulate the model over 200 years and illustrate in the same type of diagram, again starting from the old steady state in periods 0–10. Comment again with respect to your simulations being able to 'explain' empirical observations on population growth rates and economic growth rates, as those reported in Table 8.1.
4. You may have found above that decreases in the growth rate of the labour force, whether they are once and for all or gradual, do not really (in the model of semi-endogenous growth with reasonable parameters) *by themselves* produce additional growth for long enough periods to provide an explanation for the empirical relationship. Does this mean that we should dispose of the model or could there be other explanations?

PART 4 Structural Unemployment